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Modularity as an enabler for circular economy in micromobility operations

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Abstract

This thesis investigates how modular product architectures can support circular economy (CE) and profitability in micromobility Product-as-a-Service (PaaS) operations. Motivated by the significant environmental impact of current micromobility solutions, especially through Scope 3 emissions, the study examines Voi Technology's Voiager 5 (V5) e-scooter as a baseline. A conceptual modular version of the V5 was developed using Modular Function Deployment (MFD) methodology and compared against the original product through a structured case study. By redesigning the product and operations around modularity principles, including Single Minute Exchange of Dies (SMED) and Lean 8 Waste analysis, the research demonstrated potential reductions in greenhouse gas emissions, improved maintenance efficiency, and enhanced opportunities for reuse, refurbishment, and recycling. Results suggest that modularity concentrates environmental and financial impacts into replaceable modules, enabling faster service operations, extending product lifetimes, and increasing end-of-life material recovery. The findings offer practical insights for micromobility companies seeking to transition towards CE, while also highlighting critical aspect and the need for further validation through real-world pilots.

Keywords

Circular Economy, Micromobility, Modularity, Product-as-a-Service, Sustainability, Design for Circularity

Sammanfattning

Denna masteruppsats undersöker hur modulära produktarkitekturer kan förbättra resultat kopplade till cirkulär ekonomi (CE) och lönsamhet inom mikromobilitetstjänster baserade på Product-as-a-Service (PaaS). Studien motiveras av den betydande miljöpåverkan från nuvarande mikromobilitetslösningar, särskilt genom utsläpp från Scope 3. Voi Technology's elscooter Voiager 5 (V5) används som utgångspunkt. En teoretisk modulär version av V5 utvecklades med hjälp av Modular Function Deployment (MFD)-metodik och jämfördes med originalprodukten genom en strukturerad fallstudie. Genom att om designa produkten och dess drift efter modularitetsprinciper, inklusive Single Minute Exchange of Dies (SMED) och analys av LEAN:s 8 Waste, visade undersökningen på potentiella minskningar av växthusgasutsläpp, ökad effektivitet i underhållsarbetet och förbättrade möjligheter för återanvändning, renovering och återvinning. Resultaten visar att modularitet koncentrerar både miljömässiga och ekonomiska effekter till utbytbara moduler, vilket möjliggör snabbare service, förlänger produktlivslängden och ökar resursåtervinningen vid produktens livslut. Studien erbjuder praktiska insikter för mikromobilitetsföretag som strävar efter att övergå till en cirkulär ekonomi, samtidigt som behovet av vidare validering genom pilotprojekt lyfts fram.

Nyckelord

Cirkulär ekonomi, Mikromobilitet, Modularitet, Produkt som en tjänst (PaaS), Hållbarhet, Design för cirkulärekonomi

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List of acronyms and abbreviations

BOM	Bill Of Materials
CE	Circular Economy
DfMA	Design for Manufacturing and Assembly
DfD	Design for Disassembly
DSM	Design Structure Matrix
EoL	End of Life
GHG	Greenhouse Gas protocol
ISO	International Organization for Standardization
LCA	Life Cycle Analysis
MFD	Modular Function Deployment
MIM	Module Indication Matrix
PaaS	Product as a Service
PSM	Pugh Selection Matrix
QFD	Quality Function Deployment
V5	Voiager 5
V7	Voiager 7

1. Introduction

1.1. Background

Micromobility is defined by D'Acierno et. al. (2022) as small means of transport for shorter journeys, which are typically electric and designed to carry one person. They have become a central part of the urban transport solution in many cities, especially for shorter distances. Both Fan & Haper (2022) and Colmi & Polimemi (2024) have shown that most car journeys over shorter distances can be replaced with Micromobility solutions. Consequently, Micromobility services have the potential to reduce greenhouse gas (GHG) emissions caused by personal transportation and also alleviate public transportation.

However, despite its potential, there is still an environmental footprint from these services. Most of their emissions come from scope 3, which means indirect emissions in the supply chain. Voi Technologies, which is one of the major companies in the micromobility Product as a Service (PaaS) sector, reported a total of 11,237 CO2e emissions in 2022, and scope 3 contributed with 92,7% of that (Voi, 2023).

As of today, Voi operates more than 90,000 micromobility vehicles spread across 100 cities in 12 different European countries and is constantly growing and expanding. Voi (2023) expresses that one of the goals of the next generation of their vehicles is to be designed for Circular Economy (CE). Their current offer is a PaaS where Voi has ownership from crate to End-of-Life (EoL), meaning that they control the vehicles from when it gets delivered from their supplier until they are decommissioned. Several sources also claim that a PaaS business model is a good way to enable CE (Tukker & Tschner, 2006); (Laura, et al., 2015); (Yuan, et al., 2006) due to the company retaining product ownership rather than the user.

Given the high proposrtion of scope 3 emission in Voi's current operations, primarily driven from waste in their current operations and vehicle spare-part consumption, strategies that can enable an increase component lifetime, and material recovery are essential. Both CE and modularity are not only environmental strategies, but a business enabler for a PaaS mode.

To understand how CE principles can be operationalized in product design and within operations, the project outlines two common strategies within a circular economy are slowing and closing the resource loop (Mestre & Cooper, 2017). Slowing the loop strategies includes slowing the material flow in each phase of the use cycle through design for durability and product life extension (Vezzoli & Manzini, 2008), while closing the loop involves a design for recycling concept, such as enabling simple disassembly and appropriate material selections (Mestre & Cooper, 2017). These strategies align closely with the capabilities that a modular product has to offer. However, the intersection between modularity and CE strategies remains quiet unexplored. This allows for an opportunity where modularity could serve as a practical enabler for circularity for Voi.

Within the modularity literature, there is a broad consensus that modular product architecture affects how products are designed, maintained, and disposed of. Specifically, modularity allows for the separation of functions into interchangeable components, which facilitates easier disassembly, targeted repairs, and upgrades (Pahl & Beitz, 1996); (Li, et al., 2008); (Baldwin & Clark, 1997), These structural changes directly influence several enablers of CE, such as upgradeability, maintainability, reuse, and recyclability. For example, easier disassembly improves material recovery at end-of-life, while interchangeable modules can extend product lifespans through upgrades or targeted maintenance. Therefore, this thesis aims to investigate the impact that hardware modularity could have on CE within a micromobility Product-as-a-Service (PaaS) company. This will be done by analyzing Voi's current product system and operational setup and comparing it to a conceptual modular product architecture to assess potential effects on both circularity and profitability.

1.2. Objectives and research questions

The purpose of this thesis is to examine how a modular product architecture, together with a system of operations adapted for it, can influence CE potential and profitability for a micromobility PaaS company. To answer this, the following research question was formulated.

In what ways can a modular product architecture increase circularity and profitability for a micromobility Product as a Service company?

To investigate this research question, the following objectives were created to enable an analysis:

- A proposal for a modular product architecture based on the currently operated product.
- A proposal for a new system of operations based on principles for CE and modularity, adapted for a micromobility PaaS company.

These were set in regard to Vois' company operations today and were considered necessary to make a comparison.

The project has some delimitations to fit the scope for a master thesis, and due to the nature of the company:

- When looking at emissions from the product, only the material inputs' carbon footprint is considered. A full LCA of the product would be relevant but is determined beyond the project's scope.
- The study uses Vois' internal data that cannot be published due to confidentiality.
- A full modelling of the product would be relevant, but due to third-party design and manufacturing, this is not possible and beyond the scope of this project.

2. Theoretical Framework

This chapter first presents the theory behind the framework for this report and is then followed by the frameworks used within the project. The theory handles circularity, circular economy, together with implementation strategies for it together with concepts regarding modularity and the LEAN concepts 8 wastes to identify and analyze waste in the operations, and Single Minute Exchange of Dies (SMED) to support the development of a modular system that enables fast and simple module swaps.

Firstly, a definition of CE is established for this report.

2.1. Defining CE

The conventional production and consumption model today, based on take, make and dispose is called a linear model (Di Maio & Rem, 2015); (Bocken, et al., 2016). Di Mario & Rem (2015) further state that in CE models, the goal is to maintain the product or the value of the offer for as long as possible while minimizing waste. In Table 1, different definitions of CE are presented.

Table 1, Different definitions of CE.

Definition	Source
CE is a model for producing and consuming by sharing, borrowing, reusing, repairing, and recycling existing materials and products for as long as possible to create further value.	(European Parliament, 2023)
CE is a system where materials never become waste and where nature is regenerated. CE is built upon three core principles, to eliminate waste and pollution, to circulate products and materials, and to recreate nature, and that these should be driven by design.	(Ellen MacArthur Foundation, u.d.)
Implementing a circular economy involves applying multiple approaches relating to resource management in order to reduce the use and loss of resources when engaging in value creation and sharing.	ISO 59004:2024 (International Organization for Standardization, 2024)

Following the definitions presented, CE is an economic model aimed at separating industrial growth from resource consumption by extending the life of products, components, and materials. This involves maintaining the value of resources in the system for as long as possible. Rather than following a traditional linear model of "take-make-dispose," CE promotes a closed-loop system where resource input and waste, emissions, and energy leakage are minimized. Further in this report, CE is defined in the same way as the ISO standard.

When defining CE, there is also a further typology regarding some different types of standpoints within a circular offering. Shevchenko et al. (2023) defines it by the type of (I) CE loops (Closing or slowing the loop), (II) by the period of waste prevention (Products related to waste prevention in the future or in the PaaS), (III) by the type of circular strategy (Recyclable, reuseable, refurbished, remanufactured, upcycled, shared, second hand or recycled product), (IV) by the combination of circular strategies (Products contributing to either one or multiple circular strategies), (V) by the contribution to circularity potentials (Building or using circularity potentials), (VI) and by the type of circularity (Material or product circularity).

Now that a definition of CE has been made, there is also a need to understand how it can be implemented into an organization and an offering. The next section concerns some of the previous frameworks regarding the implementation of CE.

2.2. Previous frameworks for implementing CE into an organization

In the literature, a multitude of approaches for working with CE exist. Hoffmann & Pruyn (2020) mention 10R, cradle-to-cradle, butterfly diagrams and ReSOLVE as some of them, and also that multiple aspects are overlapping between them. ISO 59004:2024 (2024) also considers principles that should be considered when an organization wants to transition towards a more CE with its offering. These are shown in Table 2.

Table 2, Principles to consider for an organization transitioning towards CE from ISO 59004:2024

Principle	Explanation
Systems thinking	Consider a long-term lifecycle perspective when considering sustainability impacts.
Value creation/Sharing	To provide a shared value throughout value chain/network in a way that benefits society in a resource efficient manner.
Resource stewardship	Managing stocks and flows in a sustainable way by closing, slowing and narrowing resource flows while still ensuring availability for future generations.
Resource traceability	Collecting and maintaining data for enabling tracking of resources through their value chains and being willing to share it with interested parties.
Ecosystem resilience	Develop and implement practices and strategies to protect and contribute to the resilience and regeneration of both ecosystems and biodiversity.

Zhghenti et al. (2023) highlights organizational opportunities when adopting circular practices. First of all, it can allow companies to improve resource efficiency, generate new revenue streams, and improve their competitive position, while encouraging innovations. The study further discusses the enabling role of institutional incentives, such as subsidies and tax deductions, that can further support organizations to overcome financial and technological barriers with the implementation of circular measures. Most importantly, the development of circular capabilities allows for long-term environmental goal fulfillment and positions organization as a proactive drive in shaping a sustainable business future.

With the definitions and baselines for CE, like looking at the full product lifecycle and retaining value with the same resources, there should be somewhat of an understanding of the need to have an offer and a product that complies with CE. In the following section, the 10R framework by Cramer (2020) is presented more in-depth. The 10R framework is chosen due to its holistic viewpoint on circularity where it includes the full product lifecycle from design to EoL. Additionally, it involves multiple circular strategies ranging from smart product use to material recovery. It enables a structured approach to evaluating each stage within a product life cycle, supporting both upstream and downstream opportunities for circular improvements.

2.3. The 10R framework

In this report, the 10R framework by Cramer (2020) serves as a foundation for analyzing the impact of modularity on CE. It sets up guidelines for handling CE products and offerings. If the reader has good knowledge regarding the 10R framework for manufacturing, it is recommended that they only read the definitions in this report for each of the Rs that can be found in Table 3 and skip this chapter.

The different R strategies can be quite abstract when read in plain text, and some analogies and examples were therefore made internally for this project, which can be found in

Appendix 1. It should be noted that the 10R principles consist of two sets, one more adapted for services and business models (Potting, et al., 2017) compared to the one that is more adapted for manufacturers (Kirchherr, et al., 2017). In this thesis, the one more adapted for manufacturing is used due to the GHG emissions in scope 3 of Voi, being the GHG emissions caused by the supply chain, such as production and transportation.

In literature and this text, the Rs are grouped into three groups: smart product use and manufacturing, extended lifetime, and materials handling. Further, it is often also a hierarchy seen for the Rs, where the first ones have the most circular impact and demand a circular business model while the last ones have less impact and can be used in a linear economy. This is noted in this report, but it should also be noted that a holistic approach should be used when considering this to accomplish minimal resource usage.

2.3.1. R0-R2: Smart product use and manufacturing

R0-R2 concerns the earliest lifecycle stages for the product, at which the product is designed, developed, and conceived by the customer. Therefore, they can make a resource impact before any of the other R's and also optimize the business models for enabling them. This gives the possibility of creating a circularity transition, even before production use has started (Morseletto, 2020).

R0: Refuse

The concept of *refuse* in the context of the producer refers to as when the producer creates a norm in refusing to use hazardous materials. It further involves designing a production process where waste and the utilization of virgin materials are kept to a minimum (Bilitewski, 2012). It also concerns the usage of specific materials to make a solution more circular. Additionally, this also concerns abandoning its function or offering the same function through a different product, solution, or service (Morseletto, 2020).

In this report, refuse is defined as *a norm to avoid the use of hazardous, non-recyclable, and toxic materials within production*.

R1: Rethink

According to Reike (2024) the *rethink* concept has as a goal to create a product that is use-intensive. This could be done through sharing products or by adding additional functions to the product to give it more purpose.

In this report, rethink is defined as *reevaluating the current functionality of a product with the purpose of creating a prolonged and intensive use phase*.

R2: Reduce

The concept of *reduce* is based on the principle that products in their design should minimize the amount of material used. It is evaluated through looking at the resources used in pre-manufacturing, manufacturing, and the usage phase (Reike, 2024).

In this report, reduce is defined as *minimizing material and resource consumption through the product lifecycle*.

2.3.2. R3-R7: Extended lifetime

R3-R7 goals are to extend the lifetime of a product by letting it create value for a longer period of time with the same resources. Within them, designing for disassembly, repair, and upgradeability is crucial. This means that in order to fulfill them, a system perspective where the full lifecycle is included has to be taken into consideration during the product design (Rahman, et al., 2021).

The definition of product lifetime becomes important when utilizing R3-R7. Typically, this is defined as the time span for when a product is functional (Murakami, et al., 2010). Burn & Cooper (2010) argue that the products' lifetime should be evaluated based on the time it takes for them to become obsolete. This means that a product is obsolete when it can no longer create value. At the same time, Hollander et al. (2017) suggest that

obsolescence can be reversed and lead to further usage, which means that lifetime can be extended even further.

To reverse the obsolete state of a product, Stahel's (2010) inertia principle gives examples of how obsolescence can be avoided by only repairing products that are broken. Further, he argues that instead of recycling products, they should be remanufactured. The inertia principle also says that firms should aim to replace or treat the smallest possible parts to maintain the existing economic value of the technical system as a whole.

R3: Reuse

The concept of reuse is defined by de Brito, et al. (2003) as a product that is handed to a second consumer, and where it barely needs any adaptations to work as new. Moreover, Potting et al, (2017) present the following definition: A product that has been discarded by the first user is later re-used by a second user where it fulfils its original function.

This report defines reuse as *using the same product or component for multiple product generations or giving it a second life with the same purpose.*

R4: Repair

The concept of *repair* is according to Reike (2024) a misinterpreted strategy where it can be difficult to distinguish between repair and refurbishment. Fernández & Kekäle (2005) define repair as "bringing back to working order", while Srivastava (2008) defines it as "making it as good as new. Stahel (2010) argues that a more accurate description is "recreating its original function after minor defects".

The following definition is set for this report: *Restoring a product to working order by maintaining defects without significantly altering the original structure or functionality.*

R5: Refurbish

The concept of refurbishment requires the overall structure of a product to be intact, but single components are replaced or repaired, resulting in an overall upgrade (Fernández & Kekäle, 2005). Stahel (2010) defines refurbishment as bringing the product "up to the state-of-the-art" due to the use of newer and more advanced components.

For this report, refurbishment is defined as *making upgrades to the current system that create more value for stakeholders.*

R6: Remanufacture

Gehin et al (2008) and Lieder & Rashid (2016) define remanufactured as a full structure of multi-component product assembly that is being disassembled, checked, cleaned and where components either gets repaired or replaced when necessary. Moreover, Hollander & Bakker (2012) offer a simplified term referring to remanufacturing as reconditioning, reprocessing, or restoration. Moreover, Hollander & Bakker (2012) offer a simplified term referring to remanufacturing as reconditioning, reprocessing, or restoration.

For report this report, the definition of remanufacturing is defined as *breaking apart a product into its components, replacing what is broken, and allowing it to function as new.* In other words, remanufacturing is a more intensive form of repair that completely breaks down the product into individual components.

R7: Repurpose

The concept of *repurpose* is defined by Reike (2024) as a discarded component being adapted for another function, and therefore giving its material an entire new lifecycle and gives the analog of transforming bottles into mugs or using defect microchips for jewelers.

The definition used in this report is formulated as *giving a discarded component another function another purpose than it was intended for.*

2.3.3. R8-R9: Material handling

R8-R9 addresses the material handling in the end-of-life stage of a product. The R8-9 strategies concern an efficient handling of the products EoL to retain its initial resource value in which it would otherwise desisted for landfill or burned without heat recovery (Morseletto, 2020).

R8: Recycle

The purpose of recycling is to capture the pure material for it to be manufactured back into a functional component (Yan & Feng, 2014). This concept is not at the top within the R-strategies, as a consequence of the loss of the component's structural integrity (Graedel, et al., 2011); (Jaafar, et al., 2007).

In this report recycling is defined as *a component reaching its EoL and being processed for material recovery.*

R9: Recover

The concept of recovery is described by Potting et al (2017) as the incineration of material with energy recovery. In broad terms, recovery refers to waste that is not recycled, which instead becomes a source of energy or valuable biochemical compound (Morseletto, 2020). This aligns with the definition of how the report defines recovery.

All the definitions for the 10R framework that is used for this report can be found in Table 3.

Table 3, Summarizing the definitions used in this report for the 10R framework

R	Definition
R0-R2: Smart product use and manufacturing	
R0: Refuse	*A norm to avoid the use of hazardous, non-recyclable, and toxic material within production.*
R1: Rethink	*Revaluating the current functionality of a product with the purpose of creating a prolonged and intensive use phase.*
R2: Reduce	*Minimizing material and resource consumption through the product lifecycle.*
R3-R7: Extended lifetime	
R3: Reuse	*Using the same product or component for multiple product generations or giving it second life with the same purpose.*
R4: Repair	*Restoring a product to working order by maintaining defects without significantly altering the original structure or functionality.*
R5: Refurbish	*Making upgrades to the current system creates more value for stakeholders.*
R6: Remanufacture	*Breaking apart a product into its components, replacing what is broken and allowing it to function as new*
R7: Repurpose	*Giving a discarded component another function with another purpose than it was intended for.*
R8-R9: Material handling	
R8: Recycle	*A component reaching its end of life (EoL) and being processed for material recovery.*
R9: Recover	*The incineration of material with energy recovery.*

To implement the 10R framework within product development, companies must adopt specific circular design strategies that support and enable these R-principles throughout the product lifecycle. These design strategies act as tools that bridge the gap between abstract circular goals and practical implementation.

2.4. Circular Design Strategies

Circular design strategies ease the transition and development of products and offers to work in circularity by giving guidance for considerations within hardware and services. The three design strategies explained in this section are design for product integrity, Design for Disassembly (DfD), and Design for Recycling (DfR). In the next section, modular design is also presented. The first three strategies, Design for Product Integrity, DfD,

and DfR were chosen due to their impact on the 10Rs, and modular design was chosen due to the nature of the project. Each one of them is explained under their own headline.

2.4.1. Design for product integrity – Increase the product lifetime

Designing for product integrity means designing for durability, repairability, and upgradeability in order to resist product obsolescence (Hollander, et al., 2017). It is therefore very much in line with R3-R7 (extending lifetime). In order to create a high product integrity, Hollander argues that the following aspects have to be taken into consideration:

- **Resisting obsolescence**: Achieved with a rigid and durable design
- **Postponing obsolescence**: Enabling easy maintenance and upgradeability (also mentioned by Vezzoli & Manzini (2008))
- **Reversing obsolescence**: Through design that enables repair, refurbishment, and remanufacturing, which includes the avoidance of permanent fasteners.

Design for product integrity provides a preliminary overview of how to extend a product's lifetime, while not giving many suggestions for how it should be achieved. For that purpose, it is much better to consider design for disassembly, which the next section handles.

2.4.2. Design for disassembly

Design for Disassembly (DfD) speaks to a great extent for itself for what it concerns. According to Prioli et al. (2022), it can be evaluated by looking at collision evaluation already in a CAD state, which also can lead to more nondestructive operations and therefore increases the chance of reuse for parts.

Formentini & Ramanujan (2023) looked at the role of DfD for CE and saw that much of the DfD literature was applied in the later design stages. Further, they also found that DfD principles could have multiple impacts on CE, such as extending the technical lifetime by enabling replacements, changing the material compositions and separating materials for recycling. Within DfD, they found that the following parameters were of high importance:

- **Number of components**: How many interconnecting components are assembled in the final product
- **Number of joints, and what type of joints**: How many different joints are there and how are the joints fastened.
- **Reversibility**: What type of fastening method is utilized, for instance welding and gluing is not reversible, consequently, it constrains the ability to separate a module of components.
- **Risk of damage**: Critical and components where there is known high wear should be designed for replacement with minimal effort.

2.4.3. Design for recyclability

The design for recyclability (DfR) approaches focuses on increasing the resource efficiency by increasing the recyclability rate of a product component. The recycling rate is calculated based on proportion of recyclable materials relative to the total weight of the product. In other words, this measures the percentage of a products End of Life (EoL) materials that can be integrated back into the resource loop (Takahiro Mizuno, 2012)

DfR is the process of determining a recyclability evaluation. The process includes defining the EoL scenario for a given product by calculating its recyclability and recoverability rate and additionally analyzing material compositions. From a design perspective, this involves assessing existing product data, and identifying component types and material compositions, and determining their respective weights. Moreover,

it includes examining the product EoL through an EoL process flow model and evaluating the recyclables rates of different materials (Takahiro Mizuno, 2012).

Finally, the product design is evaluated based on two key factors being (1) whether the materials go through the process of manual sorting before or after shredding, and (2) whether the object consists of a single material or multiple materials (Takahiro Mizuno, 2012).

In the next section, modular design is presented since it is deeply connected to the research question and the aim for this thesis. Later, many of these circular design principles are connected back to modular design literature.

2.5. Modular design

As mentioned in the introduction, designing for modularity has the opportunity to increase CE and profitability for an organization. Firstly, modular design gets defined.

2.5.1. Definition of modular design.

Firstly, modular products are defined, and for that, Li et al. (2008) is used who describes modular products as machines, assemblies and components that fulfil various functions by combining modules. For modular architecture, Ulrich (1995) defines it as a one-to-one mapping between physical components and functional elements with decoupled interfaces between the components. A component in that case can also be a subassembly.

When defining a module, Lange & Imsdahl (2013) says that a description of modules often has three key characteristics: what the module contains, what the physical limit of the module is and why the module exists. Further they present the concept of modular function deployment that states that a module is a *functional building block with specified interfaces, driven by company specific reasons,* which is also stated by Erixon (1998).

To simplify and create understanding for this, a modular product and a modular product architecture consist of multiple modules. One module might have several variations depending on what the product has to accomplish, but the key is that every variation has the same standardized interface with other modules. This makes them interchangeable with each other.

2.5.2. Advantages and limitations of a modular product design

This section provides some information about the advantages and limitations that a modular product design imposes. Baldwin & Clark (1997) mentions ease of maintenance, mid-life upgrades, standardization, shortened design time and the ability to make customer specific module variations and configurations. Further, ElMaraghy & AlGeddawy (2014) shows the possibility of creating product variety with common modules to accomplish both economies of scale effects and market customization for the same product.

Earlier, DfD was introduced, and according to Li et al. (2008), disassembly planning as presented by Kuo et al. (1997) can easily be applied to a modular design. That creates a framework for how modularity can affect a product's lifecycle, containing concepts such as recycling/reuse/disposal, material selection and serviceability. It also concerns a hierarchical order for how a modular product should be assembled and disassembled.

For manufacturers, Pahl & Beitz (1996) wrote about the advantages that a modular design can provide. They mentioned simplification of scheduling and delivery dates for products, easier development, easier cost calculations, easier assembly and better lifecycle planning. Further, Erixon et. al. (1996) showed how the modular partitioning of a product allows for a parallel manufacturing flow together with a last assembly step. This allows for specialization of staff, ease of customization and of production flow.

When it comes to the downsides for the manufacturer, Pahl & Beitz (1996) mentions the additional up-front development costs for the system which also makes it harder to make fast product changes that do not align with the interfaces.

For the user, Pahl & Beitz (1996) states that the advantage of a modular system comes from shorter delivery times, better exchange and maintenance possibilities, better spare parts services and easier upgrades and changes. When it comes to the negative, they state that it is harder for manufacturers to comply with special customization that is not in the frame of the architecture and that the weight and volume of the system is higher compared to a specially designed product.

To summarize it a bit by looking at it from both a manufacturing and user standpoint, Magnusson & Pasche (2013) describes modularity as a tool to meet the customer specific demands. It is described that modularity can enable a product to keep high levels of product variety with high levels of variants while keeping complexity and its related cost to a limited level. Moreover, it means that a modular product allows for a complex structure to be decomposed. By decoupling the components and system it creates an information structure which can provide embedded coordination, ultimately enabling a company to become more strategically flexible (Simon, 1962).

In the next section, a brief overview of how a modular product architecture can be created is provided.

2.5.3. Creating a modular product architecture

A large portion of literature for how to create a modular product architecture is based on Erixon's (1996) MFD methodology (Modular Function Deployment). MFD methodology provides a step-by-step guide for the implementation while having a multiple stakeholder perspective. Erixon also extended his work in Erixon (1998) where he goes more in-depth regarding the methodology. This is further built on by Lange & Imsdahl (2013) who actively links some of the MFD steps with listening to the customer, the engineer, the manufacturing department, the quality department, the supply chain and the aftermarket. Here are the steps for the MFD process:

1. **QFD matrix (Quality Function Deployment):** clarifying the product design specifications.
2. **PSM (Pugh Selection Matrix):** Used to select and find technical solutions.
3. **MIM (Module Indication Matrix):** Used to generate concepts and identify modules and integrations.
4. Evaluate concepts and analyze interfaces with an Interface matrix and evaluation chart.
5. Improve each module with a DfMA (Design for Manufacture and Assembly).

Williamsson & Sellgren (2016) describes how clustering components into a modular architecture can significantly reduce the product complexity and decrease the risk of making design faults compared to working with traditional product development.

Further when making the module partitioning, Erixon et. al. (1996) proclaim that it should be done with module drivers as presented by (Erlandsson, et al., 1994). The main drivers mentioned there are presented in Figure 1.

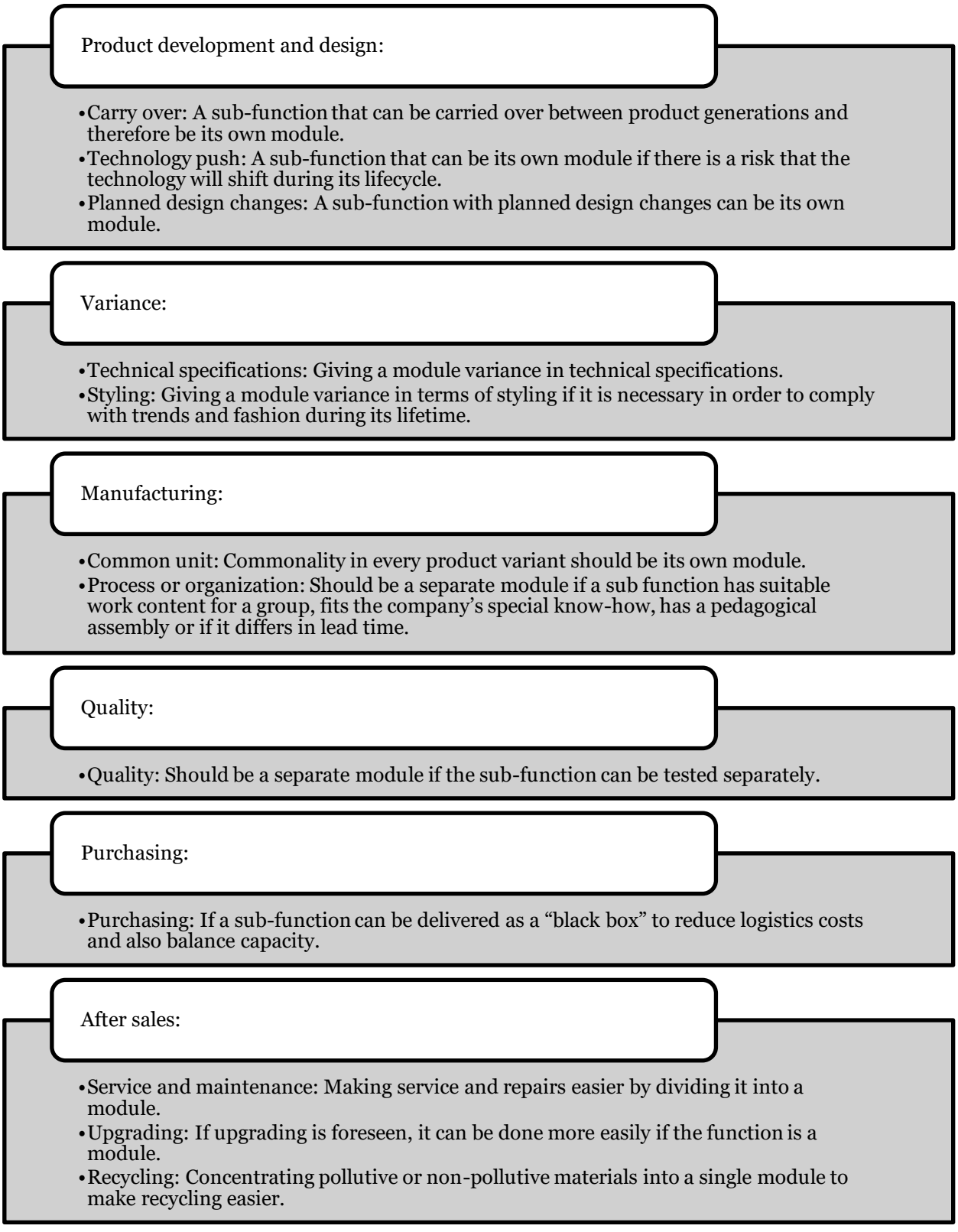


Figure 1, presentation of module drivers according to (Erlandsson, et al., 1994).

It should also be mentioned that a DSM (Design Structure Matrix) is often used together with the MFD methodology. Börjesson (2012) explains the difference between hierarchical clustering and DSM as that MFD with its hierarchical clustering is more concerned with similarity while DSM works by looking at coupling. Further explained, coupling is the terminology for technical solutions when working with DSM, and that the amount of coupling is defined by how much the technical solutions interact.

More in depth information about creating a modular product architecture can be found in Börjesson (2012), Erixon (1996), Erixon (1998) and Lange & Imsdahl (2013)

One computational method that builds upon the DSM approach is the IGTA+ algorithm (Improved Gutierrez-Thebeau Algorithm), which has been developed to automate the clustering process in modular architecture development. IGTA+ is a hill-climbing algorithm that optimizes a total-cost function by balancing intra-cluster (within-module) and extra-cluster (between-module) interactions, thereby improving module integrity and independence. Unlike traditional hierarchical clustering, IGTA+ focuses on minimizing coupling in a systematic and computationally efficient manner (Williamsson & Sellgren, 2021). Börjesson & Hölttä-Otto (2012) highlights how IGTA+ refines the clustering process by introducing enhancements such as suppressing multi-cluster allocation and using improved selection criteria during iterations. This makes IGTA+ particularly valuable when dealing with large and complex DSMs, where manual clustering would be time-consuming and unreliable.

2.5.4. Modular design for circularity

There are some links between modularity and CE that are mentioned in literature today. Mignacca et al. (2025) mentions how a modular design can improve performance in disassembly, maintainability, upgradeability, reusability and recyclability. These performance enhancements cover many of the 10R's, and the benefits are even wider. It also gives the possibility of making seamless maintenance by replacing a module instead of individual components that allows high serviceability and minimizes the product downtime. Mignacca et al. (2025) further lifts the possibility for industries with a high regard of modularization to implement reuse initiatives more easily.

Lastly before touching on the final framework, Single Minute Exchange of Dies (SMED) will be presented. This is due to its ability to work together with modularity to increase the operational efficiency of an organization by making tasks external to the product rather than internal. Building this together with Mignacca et al. (2025) has the potential to create an even greater effect.

2.6. Single Minute Exchange of Dies (SMED)

The idea of SMED is to have quick exchanges within a production line in order to minimize downtime by limiting setups to under 10 minutes. It is a part of the LEAN methodology and aims at reducing the different waste described in section 3.2. A basic principle in SMED is to move the work from internal tasks to external tasks, meaning that as much of the exchange of the parts are done while there is no downtime of the machine (Dillion & Shingo, 1985). With other words, SMED methodology intends to lower the downtime by making repairs and changes outside the product while it still can be operational.

With the literature as a basis, a developed framework for evaluating both CE and profitability will be presented.

2.7. Method for evaluating CE and profitability

Regarding literature for CE, there is often a lack of consideration for the profitability of an organization that is to implement it. It was therefore necessary to construct a method for looking at the impact in terms of value for the stakeholders that changes would affect. This is presented in the next section and is then followed by an explanation of 8 waste of LEAN (MUDA) which is used to find inefficiencies in organizations.

2.7.1. Framework for considering both CE and profitability

When literature tries to assess CE, the aspect of value impact and profitability often gets missed. There is therefore a need to develop a method and framework to see how different products and offerings affect stakeholders perceived value together with aspects of CE. This report therefore proposes a framework that takes stance in four aspects together with how the stakeholders get affected. If a similar perceived stakeholder

value cannot be presented with a CE offering, it is also hard to motivate a change in the way that business is made. Therefore, this thesis would like to argue that it is key to considering this when looking at potential changes for implementing CE.

The framework builds on the "Circular Product Design Multiple Loops diagram" developed by Mestre & Cooper (2017), which focuses on identifying design strategies for slowing and closing the resource loop across a product life cycle. However, instead of focusing on design stages, this adapted framework evaluates circularity and profitability across three stages together with other considerations. The phases evaluated are material usage over lifetime, operational affection, EoL processing, and other considerations.

These phases have been selected based on their connection to the 10R framework, profitability factors, and EoL strategies. Each phase is further partitioned into aspects which allows a stakeholder evaluation for its value effects. Affected stakeholders are identified and mapped where their impact is evaluated based on these aspects. Notably, the environment is also treated as its own stakeholder, allowing the framework to assess how the environment is influenced within each phase. The aspects are then ranked between -3 to +3 where 0 stands for no effects for each stakeholder. Finally, the mean value of each aspect is then weighed to determine the resulting impact. The framework is shown in Table 4. Figure 2 presents a step by step guide on how to use the framework.

Table 4, Stakeholder value impact framework for profitability and CE

Life cycle stages	**10R / profitability / EoL**	**Aspect**	**Stakeholder**	**…**	**Environment**	**Weighting mean**
Material usage over lifetime		[Aspect name]	[Weighting -3 to +3]			
Operational affection						
EoL Processing						
Other considerations						

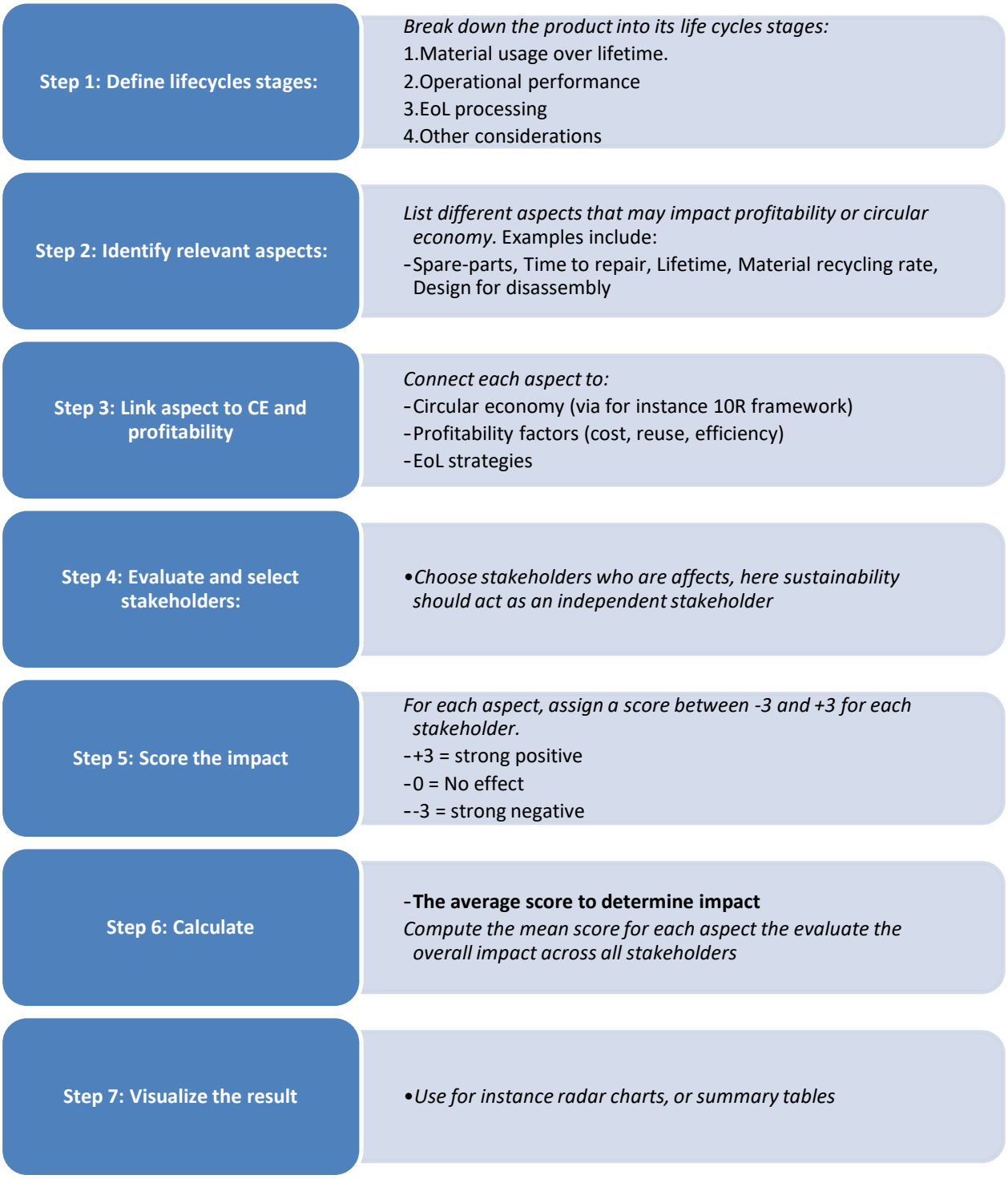


Figure 2, Description of the steps for framework considering both CE and stakeholder perceived value

Further, 8 wastes of LEAN will be presented as a tool to evaluate the inefficiency of an organization's operations.

2.7.2. The 8 wastes of LEAN – A practical approach to analyzing operations

In the context of CE, a key principle is to minimize the use of resources that are not put back into the resource loop. When analyzing a product's architecture or system, waste is the primary factor that should be removed.

To understand and analyze operation, the 8-waste method from LEAN can be used. The 8 different types of operational waste are described under the acronym TIMWOODS and are presented in Table 5 based on the explanation by Schmit (2017).

Table 5, description of the 8 wastes of LEAN.

Waste	Description
T**ransportation**	Wasted costs, resources and time when unnecessarily moving materials and products
I**nventory**	Wastes from having excess inventory of products and materials that are not processed
M**otion**	Moving people unnecessary
W**aiting**	Waste from spending time between processing steps
O**verproduction**	Waste from producing more products than what is wanted from the customers
O**verprocessing**	Making a product of higher quality than what is required by the customer
D**efects**	Waste from a service or product which does not meet customer expectations
S**kills**	Waste due to not using skills and talents of employees for value production

3. Method

This chapter describes how the research was carried out to explore how a modular product architecture can support circularity and profitability in a micromobility PaaS company. The study focuses on Voi Technologies and their Voiager 5 (V5) e-scooter. The research is based on a case study approach and builds on the theoretical framework presented earlier, especially the 10R framework, circular design strategies, and the Modular Function Deployment (MFD) method. The steps in the methodology are illustrated in Figure 3.

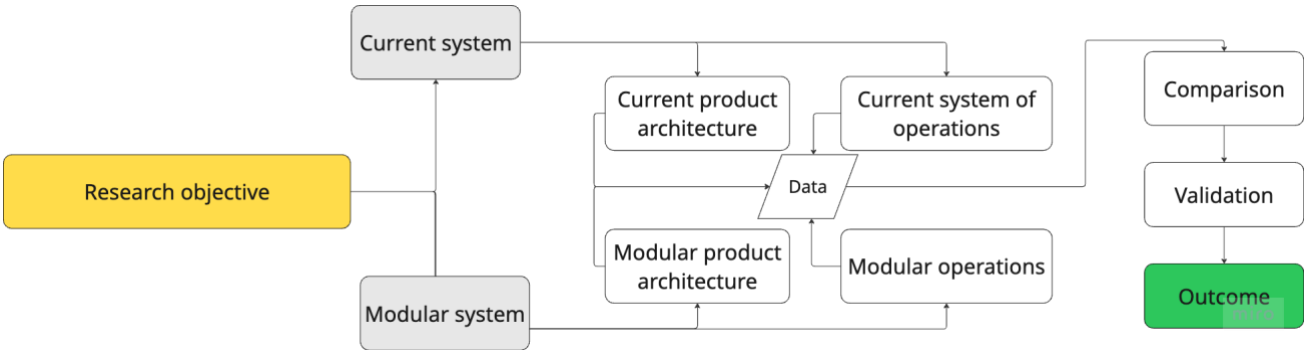


Figure 3, Overview of methods main steps

It should be noted that artificial intelligence large language models have been used to write some parts of this report. It has been used for making formulations and to correct language, but not to collect or process data.

3.1. Research Objective and Design

The aim of this study is to investigate how a modular product, along with a modular operations system, could improve both environmental and financial performance in a PaaS business.

To answer this, the study compares two scenarios: the current setup with the V5 scooter and a theoretical version of a modular scooter with adapted operations. The project takes a holistic research design approach where it combines qualitative insights from interviews and observations with quantitative data on emissions, costs and tasks. Additionally, the case study approach allows for an in-depth analysis of Vois current and conceptual modular system, while the use of established framework like 10R and MFD ensures that methodology has been incorporated to ensure the academic relevance with the project.

3.2. Current Systems Analysis

Understanding the current systems was the first step, as it provided a necessary foundation for identifying improvement opportunities. This included both studying the product architecture to understand how the scooter is built and how the different elements interact with each other. The operational process at Voi was studied to gain insight into how the product is handled through its lifecycle, from deployment, maintenance and EoL. Together, this knowledge allowed the project to find pain points within the current architecture and the operations around it.

3.2.1. Product System Architecture

The existing product architecture was studied by reviewing the Bill of Materials (BOM) for the V5 together with physically disassembling a V5 scooter. This helped identify how the product is built and how the parts within it interact. Moreover, during this process data were also gathered through a wide range of informal interviews conducted with different employees at Voi, ranging from mechanics in the warehouse, head of sustainability, chief technical officer, vice president for procurement and supply chain, fleet specialist team

leader, head of product design, and product manager. The informal interviews were held during meetings where specific questions and ideas were discussed with the purpose of gaining further insight into product maintenance, repair need, and design challenges.

To get an estimate on what each current subassembly and its inherent component has for its environmental impact on the product, an analysis was done in Ansys materials education software Granta Edu Pack which gave values for greenhouse gas emissions and potential for recycling.

3.2.2. Operations System

To get an understanding of the current operation at Voi, a field trip was done to Vois local warehouse in Stockholm. Here it could be observed how repairs are carried out on the scooters and how the practical repair stream was done. Additionally, it allowed conversations with different mechanics as well as the fleet specialist to get some other perspectives of what works well and what can be improved. To get an understanding of how Voi operations work on paper, internal documents and data were reviewed. The process for tasks like battery swaps, repairs, and scooter transport was mapped out using the collected information.

The whole operating system was first mapped on a whiteboard. Then, using the 8 wastes of LEAN, inefficiencies within the operations were identified. These findings helped when shaping the proposed modular system of operations.

3.3. Development of the Modular System

After understanding the current systems, a modular version of the product and a corresponding operations system were developed.

3.3.1. Modular Product Architecture

To develop the modular architecture, the V5 scooter was analyzed by identifying components that held a function. This was done to focus the modularization on parts that have implications on performance, and the lifecycle impact of the scooter, while excluding components such as fasteners which do not add any operational value.

Further a Design Structure Matrix (DSM) was created to map how the functional components interact in terms of geometry, mass, signal and energy. This allowed for a better understating of the system dependencies and identification of constraints when clustering the components. The DSM was then further clustered using the IGTA+ tool to support the formation of independent modules. The IGTA+ algorithm was used multiple times and after each iteration a reflection took place to validate the component clustering that formed each module.

After that, module drivers were assigned to each module based on repairs frequency, process needs, supplier strategy and recycling potential. This was done to make sure that the modules created could align with the organizational strategy, operational efficiency and CE goals.

Finally, the DSM clusters and module drivers were combined to define the final modular architecture. Interface diagrams were then created to visualize the results of how modules are connected and how they communicate between each other.

3.3.2. Modular Operations System

To integrate with the modular architecture, a new system of operations was developed in combination with Lean SMED. It was designed to reduce identified waste and make maintenance more efficient. The goal was to improve both time and resource use by going from entire scooter repairs to modular repairs. It also involves transferring scooter repairs through module exchanges to the field while doing repairs to single modules in the warehouse.

To do this, the operations were divided into field and warehouse levels. To reduce transport-related waste and scooter downtime, the field operations process was redesigned to enable in-field replacement of individual modules rather than transporting entire scooters for basic repairs. This method was implemented to minimize the environmental and operational impact of transport, reduce the risk of introducing defects during handling, and ensure quicker return-to-service times for scooters.

3.4. Comparative Analysis

To compare the original V5 products and the proposed modular architecture, the two systems were compared side-by-side. Only functional components were included to focus the analysis on how the resource distribution had changed between the modular system and the original V5.

The system was then analyzed based on four criteria: Cost share, GHG emissions, EoL potential, and GHG difference (the difference between initial GHG emissions and potential EoL recovery). Visualization was also created to further support the results by comparing the two systems using staple charts.

Moreover, to evaluate how a modular architecture supports circularity, the results were connected back to the 10R framework. The 10Rs were grouped and reviewed in three levels from R0-R2 (smart product use and manufacture), R3-R7 (Extended lifetime), and R8-R9 (material handling). Each group was analyzed in relation to the results to assess how modularity and the proposed operation around it enables circular strategies. This was done to understand modularity contribution to CE goals, and to identify which strategies are supported in the conceptual space.

The two systems were compared side-by-side. Only components with a function were included in the comparison for both the systems to ensure a fair result. The analysis looked at cost and environmental impact (GHG and recycling potential). These results were connected back to the 10R framework and helped assess whether modularity supports circularity and business goals.

3.5. Validation

To check the relevance and feasibility of the proposed modular system, a summary of the findings was presented to Voi's head of sustainability, chief technical officer, vice president of supply chain and procurement, and vice president of product. The purpose of the presentation was to understand the validity of the results from the work conducted, and if there was anything that was overseen that needed further adjustments. After the presentation there was a discussion around the work conducted where questions could be answered, and input around the work and further improvements. The feedback included mostly future work for the practical implementation of the modular system, along with some positive feedback about the conducted work.

To check the relevance and feasibility of the proposed modular system, a summary of the findings was shared with Voi's hardware responsible in a presentation. Feedback was collected regarding feasibility and potential impact. The feedback was reviewed and helped verify if the proposed system made sense in a real-world context.

3.6. Evaluating using framework for CE and profitability

To assess how a modular product architecture affects different stakeholders, the environment, and profitability, value analysis was conducted. This was done through the framework presented in section 2.5.3, where the environment was considered as its own stakeholder. The purpose was to compare how the modular and current product architecture affect circular economic principles and profitability by looking at the stakeholders perceived value.

First, aspects affected by the product architecture for each of the stages were noted and defined. These were then used to compare how it affected different stakeholders and the environment based on the proposed changes. Scoring for each aspect was performed by the project based on the qualitative and quantitative gathering from unstructured interviews during the outline of the project, and internal documentation.

Each stakeholder was weighted based on the impact from -3 (strongly negative), (same as the current), and +3 (strongly positive).

The scoring was explicitly made to reflect the context of modularity in a wider perspective, incorporating the view of external stakeholder other than Voi. The outcome was then visualized through radar charts for each stakeholder where it showed what impact modularity has across each presented aspect.

4. Results and Analysis

This chapter covers the results of the project together with an analysis. The basis of the results is built on data gathering at Voi Technology.

4.1. The product system today

Voi operates multiple models of scooters in their fleet today. This study, as stated before, concerns the Voiager 5 (V5). However, a lot of the architecture of this scooter is also similar to the other models. Currently, the lifetime of the Voiager 5 scooter is calculated to be 5 years, or 60 months for the whole scooter and 3.7 years for the batteries (Electric Avenue, 2022).

4.1.1. Current product architecture BOM

The current BOM of the V5 supplied by Ninebot is divided into 13 subassemblies. These are presented in Table 6 together with a description of what the subassembly is and what percentage of the total price it stands for. The percentage is described as Low, Medium and High due to the confidentiality of the numbers. The full BOM can be found in Appendix 2.

Table 6, Subassemblies within the Voiager V5s BOM

Name of subassembly	Description	% of end price
Stem	The handlebar part of the scooter.	Medium
Front fork and wheel	The fork together with the front wheel and the front fender.	Medium
Vertical tube	The tube connects the stem, frame and front wheel.	Low
Rear fender	The fender over the rear wheel.	Low
Rear wheel	The rear wheel of the scooter together with the motor	Medium
Rear steering lamp	The turn signals on each side of the rear fender	Low
Controller	The overall control unit of the scooter	Low
Battery	The battery that is placed within the battery compartment of the frame.	High
Taillight cable	Cable connecting controller to taillights	Low
Electronic lock	The lock that secures the battery cabin cover and the battery to the frame to prevent theft	Low
Battery cabin cover	The cover placed over the battery that integrates into the frame	Low
Kickstand	The kickstand placed on the front of the frame	Low
Frame base	The frame base that everything else is mounted on	Medium

Further, the BOM was analyzed using Granta Edu Packs function Eco Audit with the sustainability database. The battery, Internet of Things (IOT) unit and control unit were not considered for calculating the environmental impact because of uncertainties regarding the production and material composition of those parts. Table 7 presents the outcome from Granta Edu pack for each of the considered subassemblies. The material GHG% represents the total emissions in terms of GHG. Moreover, the potential EoL GHG% presents the potential resource recovery during the recycling stage. Since the actual amount of recycled content can't be guaranteed, the term "potential EoL" is used.

Table 7, Environmental contribution from different subassemblies according to Granta Edu Pack

Name of subassembly	Material GHG %	Potential EoL GHG %
Stem	25,9%	13,5%
Front fork and wheel	10,2%	11,7%
Vertical tube	6,3%	8,9%

Name of subassembly	Material GHG %	Potential EoL GHG %
Rear fender	5,1%	1,5%
Rear wheel	9,4%	9,6%
Rear steering lamp	6,1%	3,0%
Taillight cable	1,3%	1,8%
Electronic lock	2%	3,1%
Battery cabin cover	6,4%	8,1%
Kickstand	1,2%	1,6%
Frame base	27%	39%

In the next chapter, the observations made from the scooter when it was disassembled, and the inputs from those who work with service and maintenance are presented.

4.1.2. Observations from Product Disassembly and Employee Input

An in-depth analysis of the current scooter (Voiager 5) was conducted through a detailed disassembly of a decommissioned unit. This scooter had previously been removed from operation due to structural damage, specifically a bent frame and vertical tube. The disassembly process was documented systematically, beginning from the handlebars down to the rear wheel with each stage photographed to capture the interactions between components and Ninebot subassemblies. An overview of the components and their placement within the product architecture can be seen in Figure 4.



Figure 4, V5, disassembled with its major components spread out as an exploded view

A primary observation made during this process was the excessive complexity of the fastening system. Multiple screw types and sizes were identified, predominantly involving security Torx (mainly size T20) screws, combined with triangular-headed screws. This diversity in fastener significantly prolonged disassembly time, as it required frequent tool changes. Looking back at the BOM, it consists of 20 different screw types. Additionally, internal cable management presented notable difficulties. Long, interconnected cables ran from the scooter's steering column to the rear wheel and required meticulous handling to avoid damaging wires when separating different sections.

Once disassembled, the components were sorted into their respective materials so that they could be recycled. However, separating individual materials proved cumbersome and labor-intensive due to numerous fasteners and the mix of material types, highlighting potential improvements for future designs in terms of ease of recycling. There are also multiple types of plastic, aluminum and steel composition that are attached to each other and need to be separated for optimal recycling.

Refurbishment today

According to Vois internal document, the refurbishment being done today is made both in house and by external suppliers. Within the Voi organization, each local warehouse sort batteries, IOT´s, motors, controllers and heads from decommissioned Voiagers. These are then taken to one of the bigger warehouses, called Voi hubs or sent to an external supplier for refurbishment if it is deemed as possible. For example, the Voi hubs handle all the engine problems and some of the IOT issues, while external suppliers handle batteries, the rest of IOT problems and the controllers.

A presentation was held by a fleet specialist team leader who works and experiments around refurbishments. The fleet specialist showed how a motor and a drum brake could be refurbished. This was considered as an easy process that could save a lot of both economic and material value.

One view that has been expressed by multiple employees at the local warehouses that are working with repair is that their performance metrics such as completed repairs per hour and repair success rate today within Vois' own app that is used for repairs do not consider refurbishments and reuse of material. According to them, this lowers the incentive of taking the time to make refurbishments or using old parts, since this does not benefit their performance indexes.

Second life

According to Vois internal documentation, their e-scooter fleets get updates continuously year-by-year. As a result of this, older scooter models are sorted out from the operational fleet at some locations. These scooters are not always decommissioned (sent to recycling due to no longer being functional) but could either be used in another market location for Voi or be sold to an external actor where the company does not operate.

Input from employees

The fleet leaders that the group met during the project have expressed in unstructured interviews that there is an issue regarding the screws on the scooters. The concerns have been both due to the sheer mix of screws within the assembly together with the torque specifications, where many of the screwheads get damaged. This was similar to the findings that we made during our disassembly process.

Within the different operational tasks, including transporting the scooter between locations in the city and between the street and the local warehouse, designers and quality assurance staff noted that scooter often sustain damage during transport. While most of this damage is cosmetic, structural damage was also observed in some cases.

4.2. The operational system today

The current operational system at Voi Technologies consists of fleet specialists working primarily in the field and maintenance technicians operating within local warehouses. Information for this section was obtained through direct observations at Voi's warehouse facility, informal discussions with employees, and a detailed review of internal documents, procedures, and operational data provided by the company. Each of the systems contains its own apps, where the fleet specialists' complete tasks, and the maintenance crew does repairs and quality checks.

4.2.1. Description of tasks for fleet specialists as operated today

Fleet specialists manage daily field operations using a mobile application that coordinates various tasks. These tasks include battery swaps, rebalancing scooter distribution, quality checks, transportation of scooters for repair, scooter deployment, and minor repairs conducted on-site. Based on internal data from Voi, battery swapping emerges as the most frequent and costly, assuming one completed task costs the same amount for each different task done. Operation, representing approximately 57% of the total tasks done. Rebalancing scooters, or moving them to predefined optimal locations, constitutes around 19% of total costs. Other operations such as scooter transportation, deployment, and in field quality checks account for smaller shares

but are critical to maintaining fleet availability and operational efficiency. The operation of the company is described by Figure 5 which illustrates how different tasks are done. The distribution of tasks and what they are meant to address are shown in table 8 together with how many percent of the tasks cost they stand for. Due to the confidentiality of the numbers, the parts are presented as High, Medium and Low.

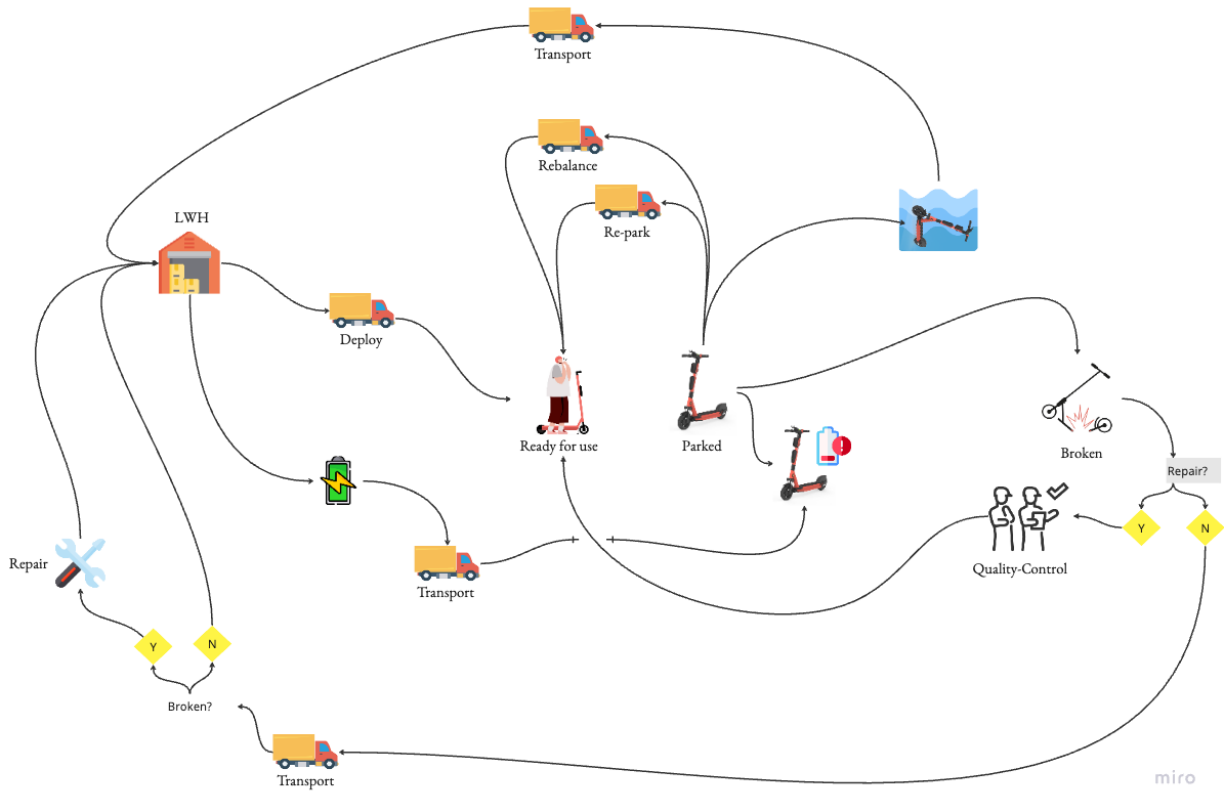


Figure 5, Flowchart of Voi's current operations

Table 8, description of tasks and how much of the task cost they contribute to

Name of task	Description of task	Part of total cost
Battery swap	Change of a scooter discharged battery with a fully charged one	High
Rebalance	Relocation of a scooter to a prescribed node	High
In field quality check	Checking wherever a scooter is broken or if it needs repair/maintenance.	Low
Transport	Moving scooters to VOI warehouse	Low
Deploy	Transport of a vehicle from a VOI warehouse to a prescribed node	Low
Repark	Repositioning a scooter according to the parking rules and/or following a provided instruction	Low
In field repair	Repairing components or adjusting but not bringing the scooter back to the warehouse.	Low
Rescue	Rescuing a scooter from water or from unauthorized possession	Low
License plate swap	Switching the scooters license plate	Low
Storage	Putting a scooter in storage due to change in demand	Low

In figure 6, the decision flow for tasks is shown for their fleet specialists' operations today. This flowchart tells the fleet specialists and the app what action each individual scooter needs in order to perform optimally.

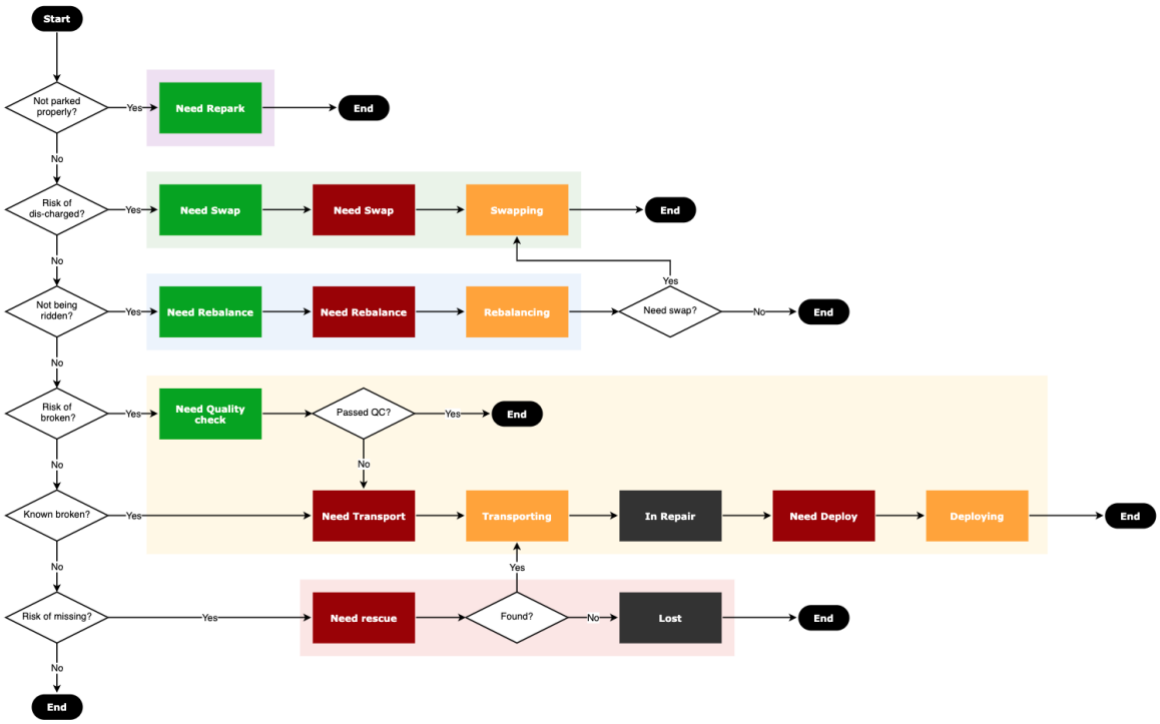


Figure 6, description of the task operation

4.2.2. Description of operations for local warehouses

Local warehouse operations handle repairs, refurbishments, quality assurance checks, and recycling activities. When scooters requiring maintenance arrive at the warehouse, they undergo an initial diagnostic evaluation. This evaluation determines whether the scooter is repairable or if it must be decommissioned due to substantial structural damage, such as severe deformation of the scooter frame. Repairable scooters are categorized based on the complexity of required repairs, either as “easy” or “hard” fixes. Subsequently, these scooters proceed to designated repair stations for detailed service. Figure 7 contains a simplified description of their flow.



Figure 7, description of flow in local warehouses

Following repairs, scooters undergo quality checks to verify functionality and safety compliance before being passed over to the field operations team. Scooters failing these quality checks are cycled back into the repair process until they meet operational standards. If it passes, it gets sent to deployment with the fleet specialists, and if it doesn’t pass it gets sent back to repair. Table 9 shows the 20 most common repairs, but as can be seen, most of the repairs fits under the repair activity “other”. On average, a scooter gets repaired around four times per year. The parts that cost the most in terms of spare parts each year are the motors, IOTs, and the controllers. Due to the confidentiality of the numbers regarding repair activities, they are exchanged into High, Medium and Low.

Table 9, Most common repairs by repair activity

Repair Activity	Proportion of total
Rear drum brake	Medium
Loose neck	Medium
Scooter off	Low
Dirty	Low
Electrical troubleshooting	Low
Steering column replacement	Low
Latch retrofit scooter/controller	Low
Brake tightening L	Low
Front wheel bearing	Low
Noisy engine/ vibrations	Low
Null	Low
Brake tightening R	Low
Battery latch stuck	Low
Front drum brake	Low
Steering bearing	Low
Scooter unresponsive	Low
Brake cable damaged rear	Low
Voi stickers	Low
Dashboard assembly	Low
Controller	Low
Other	High

Regarding the repairs of components, there is not a known number since all of the repairs are not tracked centrally. But from what is shown in the statistics and internal documentation, it is profitable for local warehouses to refurbish and that the work done locally is considered cost savings. There is also a pilot that just started in parallel with this project on a higher level of refurbishment and remanufacturing.

4.2.3. Waste analysis of the current operations

To analyze the current operations practices at Voi today, an 8 Waste analysis was done on the scooter's operations. Today the tasks done in the field involve multiple tasks that do not bring any value for Voi or the end customer. These tasks were identified as waste where there were movements of both product and employees which were deemed unnecessary. Additionally, the current operations include mechanics with a lot of competence performing simple maintenance tasks. The full breakdown of the analysis can be seen in Appendix 3.

This section is followed by a proposal for a new product architecture, together with an adapted system of operations intended to minimize waste.

4.3. The product, modularized

When developing the modularized product, the V5 scooter was analyzed by considering only the components that provided a specific functional role or purpose within the scooter. Functional parts are those that directly contribute to the scooter's performance or operation, such as the steering column that enables steering and provides structural support. Conversely, components like screws were categorized as non-functional due to their passive fastening role, which does not actively contribute to the scooter's functionality. The chosen parts can be found in Appendix 4.

4.3.1. DSM clustering

A top-down analytical approach was applied, starting with the identification of functional components that provide distinct product properties within the V5 today. These components were organized into a Design Structure Matrix (DSM), detailing their interactions by identifying interfaces for Geometry (G), Mass translation (M), Signal (S), and Energy (E). Using the IGTA+ tool developed by Thebeau (2001), the DSM was then clustered to form modular groups. The resulting clusters, detailed in Appendix 4, guided the development of the final modular architecture in conjunction with modular drivers.

4.3.2. Module driver assignment

The assignment of module drivers was made considering the repairs and GHG emissions that were found in section 4.1. When assigning module drivers, a brainstorming was held in order to reflect on the choices. The selected module drivers are defined and motivated in this section:

Common unit were assigned universally to all technical solutions, indicating no variation among scooters.

Serviceability was assigned to the front and rear brakes, brake levers, IoT device, motor, and wheels due to their frequent need for maintenance, as observed in the V5 maintenance records.

Process/Organization was applied to the front and rear wheels because servicing these components necessitates specialized equipment, such as wheel presses, not available at all local warehouses. This driver was also assigned to motors and brakes due to internal process efficiencies.

Separate testing was designated for electronic control components, including the IoT device, control unit, battery lock, and battery, due to their critical nature in scooter functionality.

Strategic supplier and upgrading were given to the battery because it is sourced externally from specialized partners and represents a key opportunity for future upgrades.

Recycling was attributed to the frame and vertical tube for their high end-of-life (EoL) recycling potential. Electronics and lighting components also received this driver due to recycling complexities, while the fork and hook were included due to favorable recycling rates.

A detailed table of the components and their corresponding module drivers is available in Appendix 4.

4.3.3. Module creation

The modular structure emerged clearly from the DSM and module driver analysis. Interfaces between modules were categorized into three types: Attachment (A) for physical connections, Transfer (T) for movement of material or energy between modules, and Command/Control (C) for operational signals. Figure 8 illustrates these module interfaces.

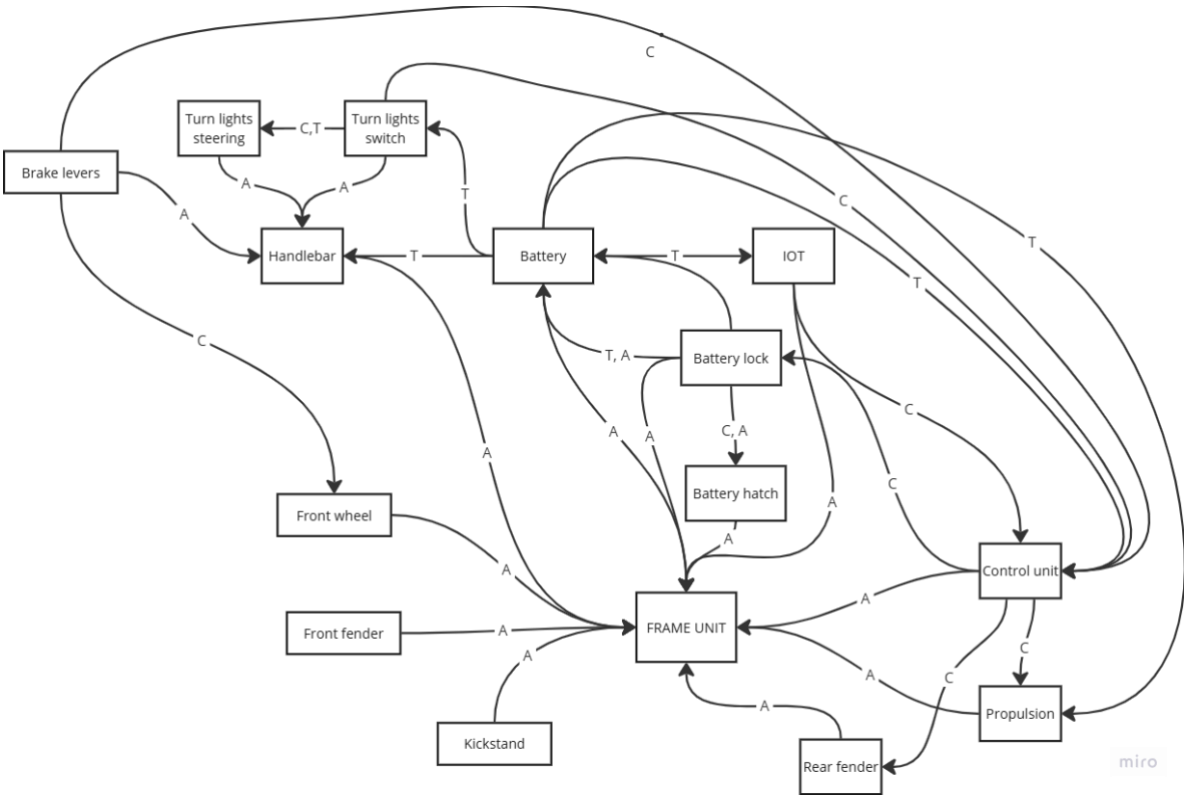


Figure 8, Interface diagram between modules with denotation for type of interface (Attachment, Control or Transfer)

The majority of modules connect directly to the frame, forming a hybrid bus-hamburger assembly configuration. Further interface refinements, including pathways for controls and transfers, are depicted in Figure 9. By leveraging the frame and handlebar as central hubs for command-and-control signals, individual modules become significantly more independent, simplifying their disassembly and maintenance. This also enables SMED for the external modules, since they are limited to only having interfaces to one other module.

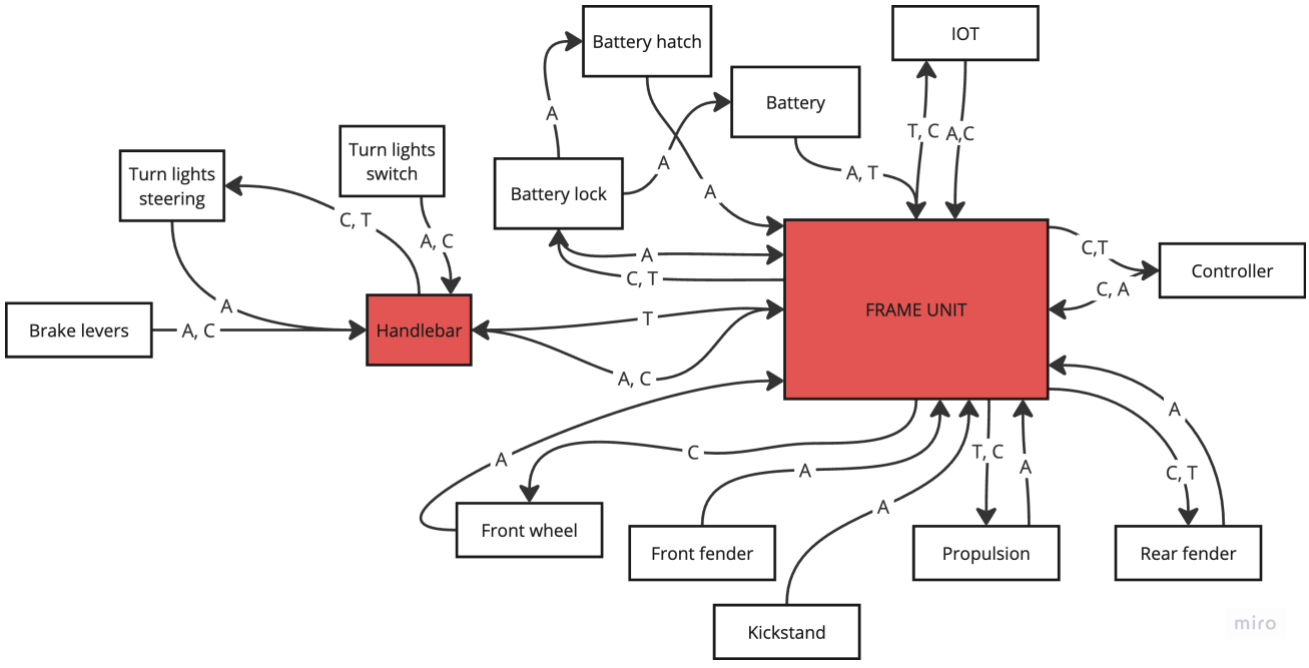


Figure 9, interface diagram with controls and transfers Passing through modules.

Further, an interface diagram including pictures of the modules can be viewed in Figure 10.

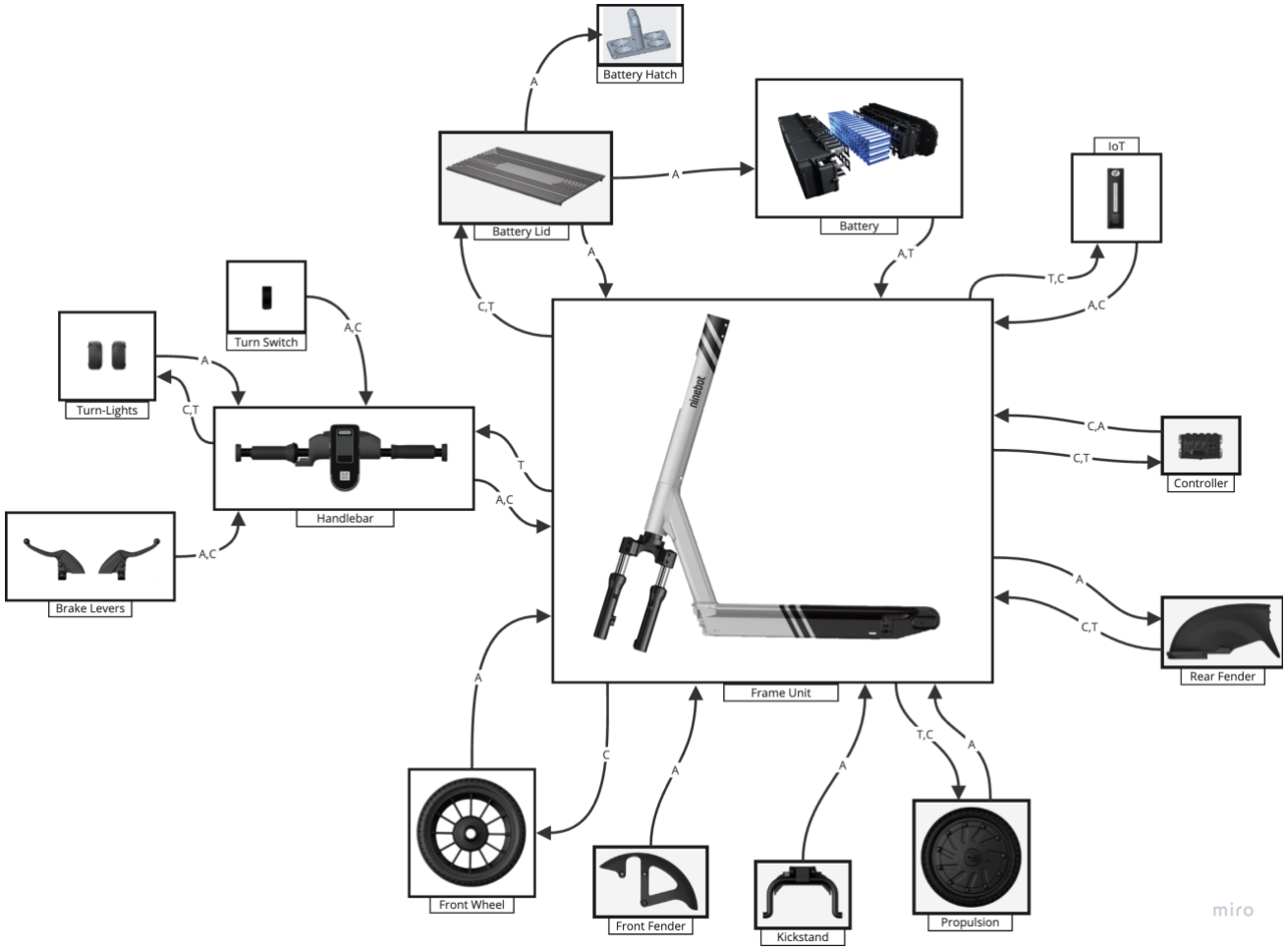


Figure 10, Interface diagram with pictures of modules

Further, an analysis of the rearrangement of the BOM is presented, together with its new values for price distribution, GHG and potential EoL.

4.3.4. BOM for the modularized product architecture

Table 10 provides a detailed breakdown of the modularized Bill of Materials (BOM), presenting costs, greenhouse gas (GHG) emissions, potential end-of-life (EoL) recovery, and GHG emission differences between the GHG emission from manufacturing and EoL. It should be noted that the modularized BOM cannot be compared to the original V5 BOM in Table 6 since this one does not consider the battery, controller, IOT and the non-functional components that are all included there. Due to the confidentiality of the numbers regarding costs, it is shown as High, Medium and Low instead.

Table 10, Modularized BOM, Costs, GHG, potential EoL GHG and GHG difference

Module	Cost %	GHG%	EoL%	GHG Diff %
Handlebar	Medium	21,47%	26,20%	17,61%
Turn lights steering	Low	0,44%	0,29%	0,55%
Brake levers	Low	0,78%	1,16%	0,47%
Turn lights switch	Low	0,46%	0,23%	0,65%
Frame unit	High	31,46%	38,88%	25,41%
Front wheel	Low	2,47%	1,92%	2,92%
Kickstand	Low	10,30%	13,84%	7,43%
Rear fender	Low	14,51%	1,82%	24,83%
Propulsion	Medium	6,40%	8,40%	4,77%
Control unit	Medium	4,65%	0,00%	8,43%
Battery lock	Low	3,01%	1,81%	3,98%
Battery hatch	Low	4,06%	5,45%	2,93%

4.4. The operation of a modularized product

The following section describes the proposed operational system designed specifically to complement the modular product architecture presented in section 4.3. This operational system builds upon the principles of modularity, the 10R framework, and that SMED-LEAN methodologies would be applicable for the modular system. The primary objective of this new system is to enhance efficiency, reduce waste, and increase circularity.

4.4.1. Field Operation

Within section 4.2.3 it was found that the current operations produce waste by moving the product for repairs. By applying a modularized product architecture, in combination with designs for SMED, the possibility to change modules in-field emerges. This means that instead of transporting a scooter from the city to the local warehouse for repairs, only the module containing the error can be replaced. This allows for a revision of the chain of decisions shown in Figure 5. The new, proposed chain of decisions is shown in Figure 11, notably only the affected change is displayed in the Figure 11, for the full operations, see Appendix 5.

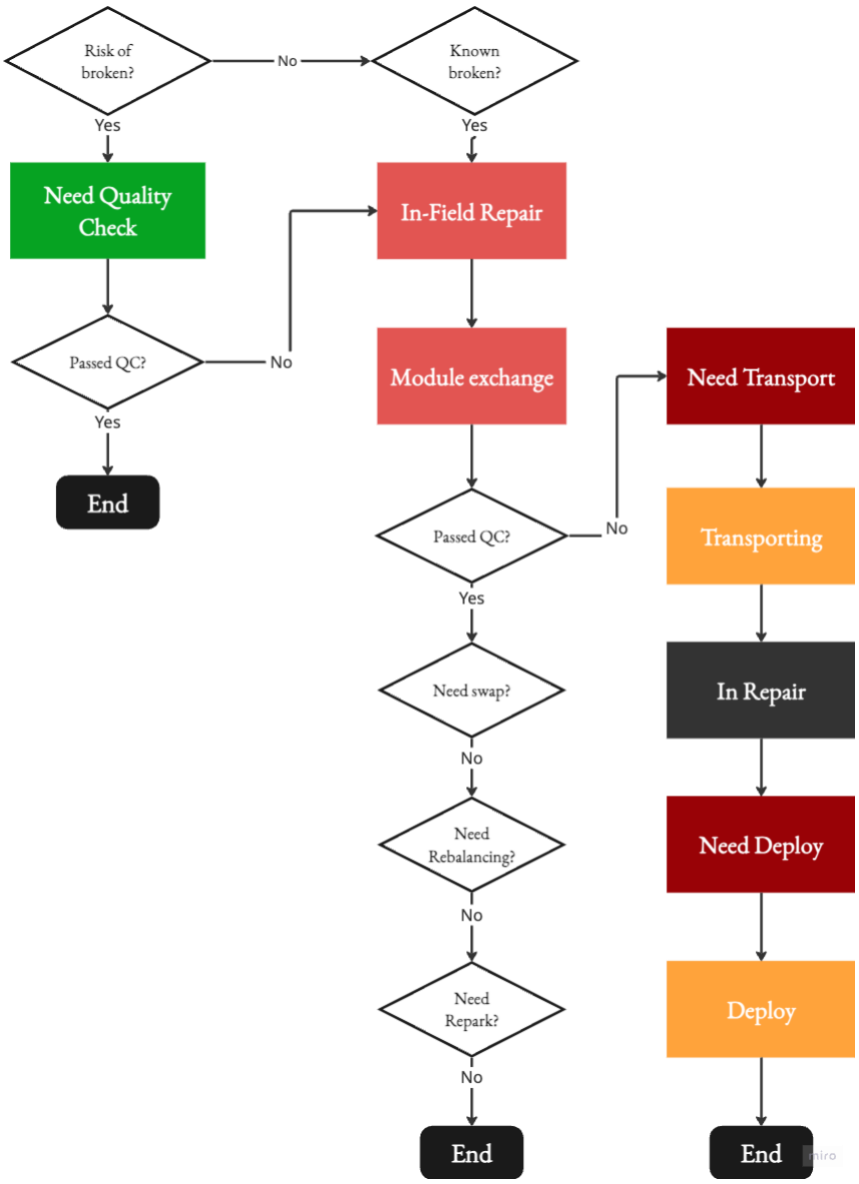


Figure 11, Field operation chain, with the incorporation of modularity

By making in-field repairs, the transport waste both to and from the local warehouses can be reduced. In turn, this should also reduce the waste of defects that come from transport as mentioned in section 4.1.2. Further, this also decreases the downtime of the scooter. Only remanufactured or broken modules need to be transported which are also easier to pack within the vans due to them not being as bulky as an entire scooter.

4.4.2. Warehouse Operation

Applying a modular system would allow for applying the parallel workflow as proposed by Erixon et. al. (1996). But instead of manufacturing, it could be applied for repairs, remanufacturing and refurbishments. So instead of considering an entire scooter when repairs are made, a single module can be considered. This allows for a setup of stations that are adapted for single modules rather than for the entire scooter. Further, less experienced employees can do the work since they only need to know the standard operating procedure, a single module. It also increases the efficiency of the warehouse operations by making the work more standardized since there is a single way to do all repairs. The overall workflow is presented in Figure 12.

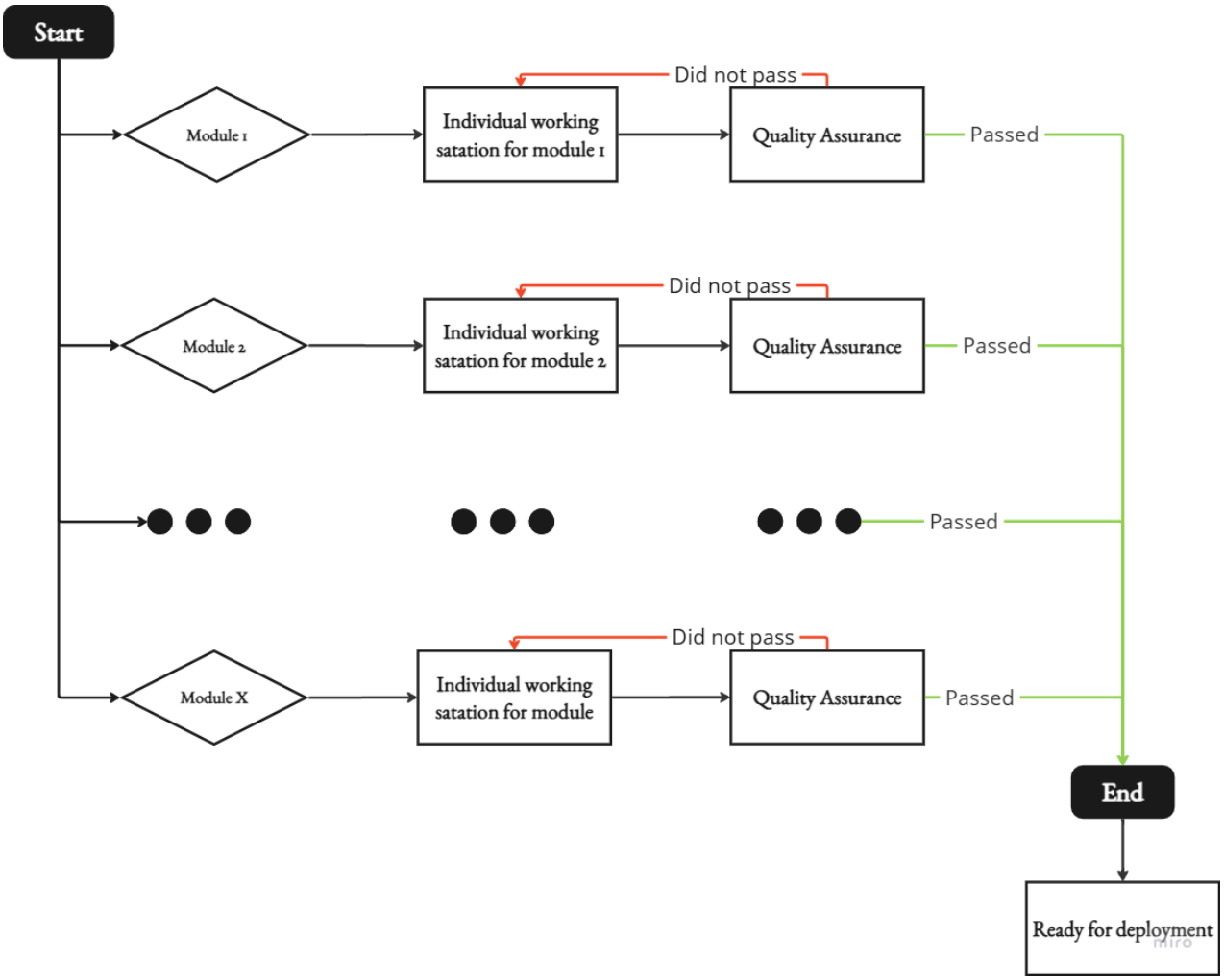


Figure 12, visualization of proposed workflow for warehouse operations on a modular level.

4.4.3. Modular repairs/remanufacturing

The workflow activities proposed for module repairs in the warehouse are as follows:

- A module enters the warehouse where it is sorted and sent to its workstation. Once there, it is initially quality checked. If the module is determined to be broken it is sent to its respective module section and set to “in repair” state.
- The module is broken down and cleaned. Each part is checked based on its initial requirements to check that it still can fulfill its intended function within the module. If the part Passes, it is made available for reassembly, if not it is sent to recycling. This can be done for multiple modules in one flow.
- If enough parts pass, they are ready to be reassembled. If there is a lack of parts, new ones are used to complete the modules.
- The new module gets quality checked based on its module functions. If it passes, it is set to ready for usage, if not, the error is looked over and seen if it can be fixed. If not, the module is disassembled and the parts that are not broken are sent to the module station to be reused for future repairs.

4.4.4. Modular refurbishments

When a module is set to have an upgrade due to the need of new design or new functionality, it is set to happen within the modular repair flow. Instead of decommissioning the entire product, only the components that are needed to be exchanged within the module are changed.

In Figure 13, a proposal for the workflow at individual module's workstation is shown that incorporates both the repair/remanufacture of a module and the refurbishment process.

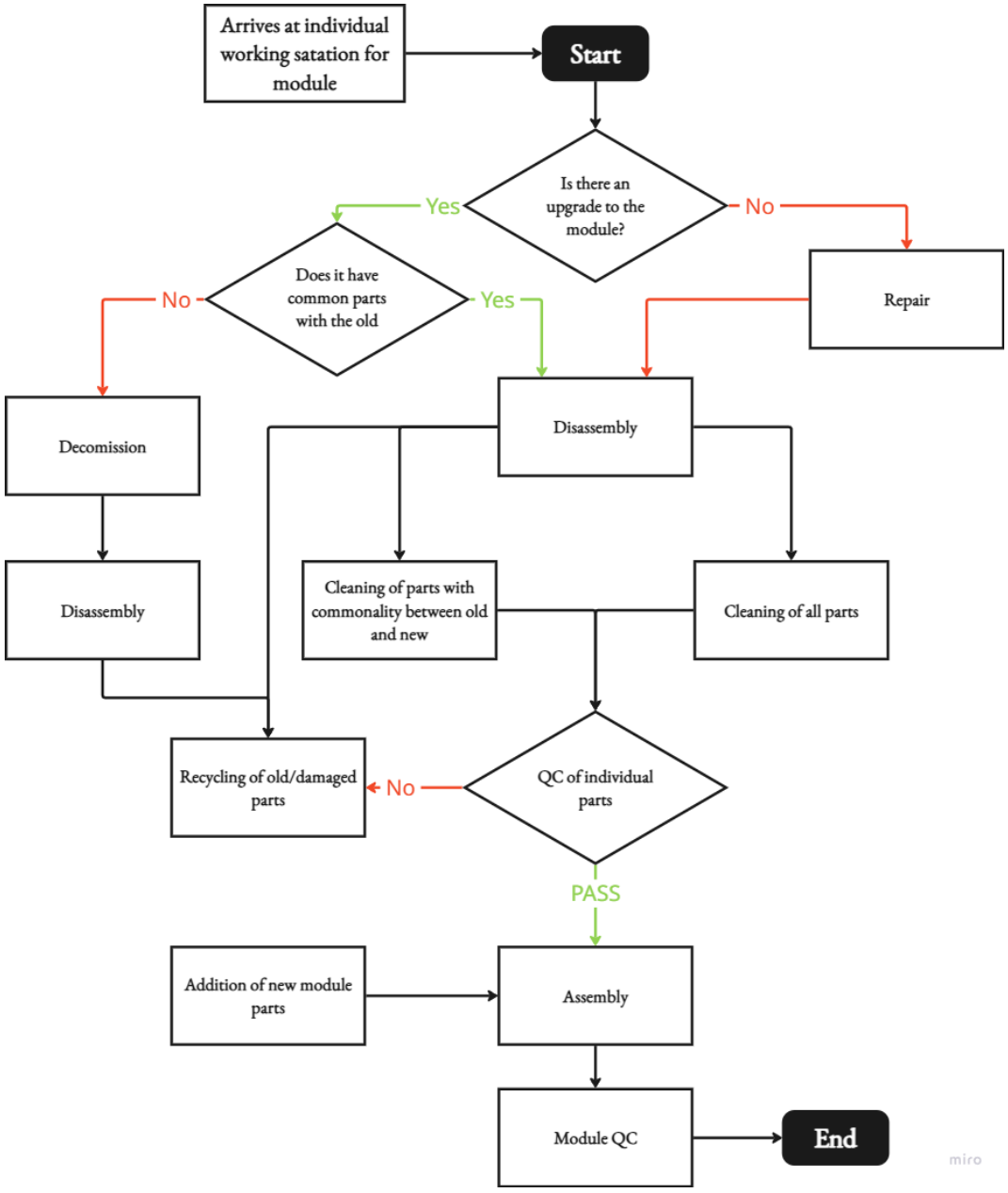


Figure 13, proposal for workflow at an individual module's workstation

4.5. Comparison between the V5 product systems and the proposed modular product systems BOMs

This section compares Voi's existing V5 scooter together with its product architecture and operational setup with the proposed modular product architecture and its corresponding operations.

4.5.1. Comparison of the BOMs of the V5 and the proposed modular product architecture

In this section, it is important to denote that the comparison is made by only looking at the functional components. This means that the information previously presented for the V5 in Table 6 and Table 7 are not the same as the information now presented for the V5, since that does not include non-functional components. This is done in order to look at the distribution of resources, rather than the total emission and total cost of the product. The subassemblies for the analysis of the V5 BOM are presented in Table 11 and the analysis of the BOM for the proposed modular architecture is provided in Table 12. The number in the parenthesis in each row corresponds to the staple in the diagrams that are later presented.

Table 11, Analysis of the V5 BOM considering only the component with a technical function from the modularization.

Original subassembly	Cost %	GHG%	EoL%	GHG Diff %
Battery cabin cover	Low	4,07% (4)	5,45% (6)	2,94% (3)
Controller	Low	4,66% (5)	0,00% (1)	8,47% (7)
Electronic lock	Low	3,01% (2)	1,81% (4)	4,00% (4)
Frame base	High	18,50% (10)	20,98% (10)	16,46% (9)
Front fork and wheel	High	13,67% (9)	17,85% (9)	10,25% (8)
Kickstand	Low	8,07% (7)	10,55% (8)	6,04% (6)
Rear fender	Low	12,89% (8)	0,13% (2)	23,32% (11)
Rear steering lamp	Low	1,43% (1)	1,69% (3)	1,22% (1)
Rear wheel	High	6,41% (6)	8,40% (7)	4,79% (5)
Stem	High	23,59% (11)	27,90% (11)	20,08% (10)
Vertical tube	Low	3,70% (3)	5,25% (5)	2,43% (2)

Table 12, Analysis of the BOM from the components of the proposed modular architecture

Modules	Cost %	GHG%	EoL%	GHG Diff %
Handlebar	Medium	21,47% (10)	26,20% (10)	17,61% (9)
Turn lights steering	Low	0,44% (1)	0,29% (2)	0,55% (2)
Brake levers	Low	0,78% (2)	1,16% (3)	0,47% (1)
Frame unit	High	31,46% (11)	38,88% (11)	25,41% (11)
Front wheel	Low	2,47% (3)	1,92% (6)	2,92% (3)
Kickstand	Low	10,30% (8)	13,84% (9)	7,43% (7)
Rear fender	Low	14,51% (9)	1,82% (5)	24,83% (10)
Propulsion	Medium	6,40% (7)	8,40% (8)	4,77% (6)
Control unit	Medium	4,65% (6)	0,00% (1)	8,43% (8)
Battery lock	Low	3,01% (4)	1,81% (4)	3,98% (5)
Battery hatch	Low	4,06% (5)	5,45% (7)	2,93% (4)

To visualize the data, diagrams were made. The number in the diagram's staples can be looked up in the tables corresponding column. The first diagram, Figure 14, shows the distribution of cost where it is clear that some modules carries a heavier part of the cost.

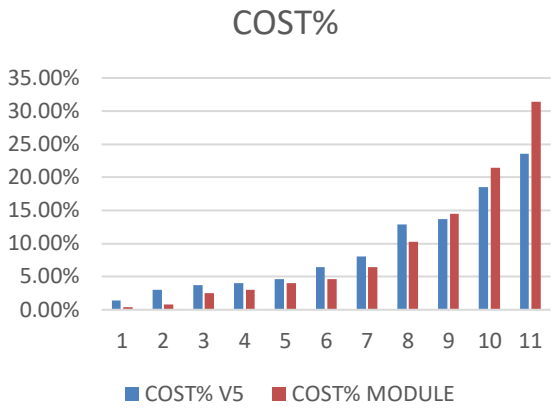


Figure 14, visualizing the distribution of cost between the partition of the V5 and the modularized product.

In Figure 15, the distribution of the ingoing material GHG is shown between the modularized product and the V5. Due to the confidentiality of the information, which module that represents which bar cannot be presented.

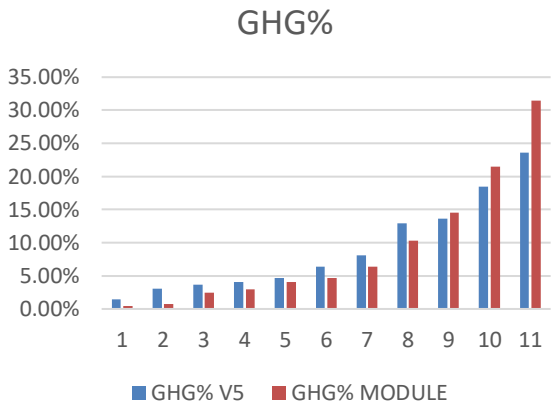


Figure 15, visualizing the distribution of GHG from the materials between the partition of the V5 and the modularized product.

Further, in Figure 16, the partition of the potential EoL is shown between the modules and the V5 scooter. Here, it is also shown that some of the partitions like handlebar (10) and frame unit (11) contain a larger part of the potential EoL GHG in the modularized product compared to the original V5.

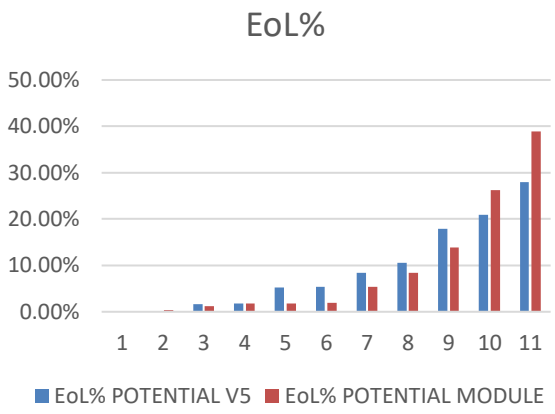


Figure 16, visualizing the distribution of potential EoL between the original V5 and the modularized product.

Figure 17 shows the distribution of GHG difference between the V5 and the modularized product. This is calculated by adding the material GHG and the potential EoL. It is shown that some of the modularized partitions most notably, frame base (11), handlebar (10) and kickstand (9).

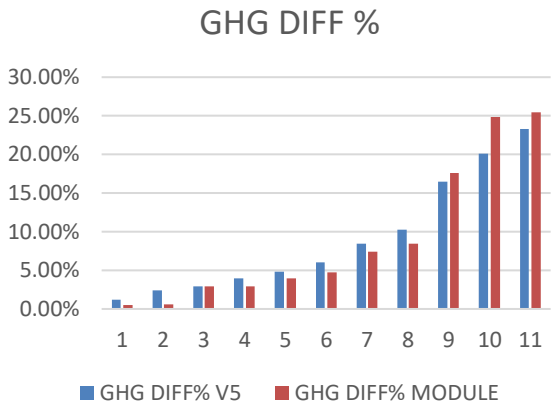


Figure 17, visualizing the distribution of GHG difference between the original V5 and the modularized product.

4.6. Analysis of implications of modularity for CE and profitability

This section analyses what the results implies that modularity has on CE and profitability for a micromobility PaaS company. It is presented by first looking at the 10Rs, followed by profitability and lastly by looking at other implications and considerations that have been found during the project.

4.6.1. Implications on CE by looking at the 10R for micromobility PaaS and modularity

This is presented according to the grouping of the 10Rs R0-R2, R3-R7 and R8-R9.

R0-R2

R0: Refuse does not seem to be addressed by modularity since the creation of the product architecture does not really consider material selections. Regarding material usage, section 4.5 shows that modularity alone can provide a shift in resource distribution within a products partition. In turn, this can help design certain modules to be more use intensive and therefore help with **R1: Rethink**. Further, a modular architecture can

enable mid-life upgrades by allowing for modular upgrades to the scooter, instead of buying a new one, this further can extend the lifetime of the product and therefore also enhance R1 together with **R2: Reduce.** It can be argued that modules could be exchanged whenever a small single component has caused it to become defective. However, with an operational system that includes processes for separating and reassembling modules, the components within each module can be continuously reused to build new, functional modules. Since only the module that has a wanted upgrade needs to be changed, it should still fit the rest of the modular product architecture. Regarding **R0: Refuse,** modularity does not seem to influence it at all, but it is rather a question of using other design principles.

R3-R7

Within the R3-R7 level is where modularity can have the biggest impact. By applying modularity together with the proposed system of rigorous disassembly, the possibility to extend the average lifetime of all the resources emerges. By not looking at the scooter as a whole product with its own lifetime, but instead breaking it down into modules, the complexity gets lowered (Magnusson & Pasche, 2013) and customer experiences and values can be altered.

A modular product platform allows working with a long-term lifecycle perspective as mentioned by ISO 59004:2024. Further, it allows for **R3: Reuse** where modules can be quickly exchanged thanks to SMED and good interfaces. Reuse also gets used by doing mid-lifetime upgrades to the configurations. **R4: Repair** also gets easier with modularity due to the fact that only a faulty partition of the scooter gets handled instead of the entire product. **R5: Refurbish** is also positively affected, since only the module that has a function upgrade has to be processed in order to get a new function into it. **R6: Remanufacture** is handled to a big extent by the workflow presented in Figure 13 within chapter 4.4. Further, when it comes to **R7: Repurpose**, the modular product architecture as presented in this report does not contribute to any further value.

R8-R9

In this report, it is shown in section 4.5.1 that modularity can contribute to collecting materials and potential EoL resources to specific modules, and therefore also contribute to **R8: Recycle.** Furthermore, looking at **R9: Recovery**, this work has not found any links between modularity and recovery.

4.6.2. Implications for profitability

This work presents multiple implications regarding how modularity can affect the company's profitability for both operations and hardware procurement. By clustering repairs and maintenance around specific modules, rather than overhauling an entire scooter, there is potential to eliminate or reduce certain tasks entirely. This approach aligns with LEAN principles both in-field and at local warehouses: repair times are shortened, fewer scooters require full removal from circulation, and technicians can focus their skills where they add the most value. Such efficiency gains become even more pronounced in combination with SMED, because only the affected module is swapped or serviced, maintaining high utilization rates and reducing downtime. Additionally, by keeping the scooter within the field it also reduces the stationary inventory within the local warehouses and maintains the scooter where it creates value for Voi.

Moreover, concentrating costs and resources into clearly defined modules emphasizes the importance of extending each module's useful life. If a particularly high-value module (for example, the battery or the motor) can be refurbished or upgraded instead of replaced, the original investment is amortized over a longer period, ultimately reducing total cost of ownership. While there may be some initial trade-offs in development and manufacturing, such as ensuring robust interfaces, these added upfront costs are generally offset by lower maintenance expenses and less frequent full-scooter replacements over time.

The procurement and ongoing upgrade of new scooters also benefit from modularity, since core components can be refreshed without discarding the base structure. As emerging technologies become available, operators can retrofit older scooters with improved modules, retaining much of the initially invested material and reducing large capital expenditures. This adaptive approach enhances profitability by extending revenue-generating lifespans, lowering warehousing and parts inventory demands, and enabling the fleet to stay competitive without continually buying entire new units.

4.6.3. Other implications and considerations found

Modular architecture can also streamline iterative development and reduce the complexity of large-scale design overhauls similarity to what Simon (1962) discusses. By isolating functionality within distinct modules, new technology or more efficient components can be introduced quickly, avoiding disruptive, full-system redesigns and allowing the fleet to adapt to evolving market demands. This is further support by Baldwin & Clark (1997) and Mignacca et al (2025) who highlight modularity role in facilitating upgrades and adaptability. In some cases, these upgrades might even open opportunities for secondary revenue streams, such as selling refurbished or overhauled modules to other operators or using them for pilot programs in new regions.

Another positive consequence is the potential for a simpler supply chain, since the focus shifts from an array of spare parts to the procurement of larger, standardized modules, which can often be purchased at scale and tested independently for consistent quality. This is also supported by Pahl & Beitz (1996) and ElMaraghy & AlGeddawy (2014) who talks about how modularity can simplify logistics, create economies of scale and allow for easier lifecycle planning.

However, certain drawbacks must also be considered. Developing a thoroughly modular product often comes with higher upfront costs, both for engineering resources and component tooling. Additional attention is required in defining robust module interfaces, which can be complex when dealing with multiple fasteners, cable routing, or safety-critical functions. This insight further aligns with Pahl & Beitz (1996) that discuss the increase in up-front costs and complexity within modular interfaces. Over-partitioning of the scooter may also lead to redundant material usage or added weight, which could undermine durability and user experience. There is a risk, too, that suppliers who provide specialized modules could have more influence over timelines and prices, potentially driving up costs if strong partnerships and contractual safeguards are not in place. For operators, an overly granular module strategy might yield diminishing returns if certain modules rarely fail or are inexpensive to replace in the first place. Decisions on where to draw the boundaries of each module must therefore be made carefully, taking both operational data and strategic cost analyses into account so that the long-term benefits of modularity can outweigh these potential drawbacks.

Further this work also relies on SMED in order to have the desired operations system. This is something that is not currently investigated wherever it would work for some of the safety-critical systems. For example, having interfaces for a brake wire could have implications for the safety and the performance of that system. This also ties back to the principles of DfD where Formentini & Ramanujan (2023) mentions that safety should be accounted for as a key consideration. Consequently, for the accurate of SMED and fast module swaps to work in the practical sense the safety aspect must be held up to the same standard as without it, following the principles of DfD.

One other consideration that should be made when modularizing a product for a micromobility PaaS company is also to look at parts commonality in order to lower the number of different components within its stock. This is also what ElMaraghy & AlGeddawy (2014) discussed where they mention how common modules between a product platform allows for an increase in the operational efficiency and also reduced product complexity. For Vois example, looking at what commonality can be achieved between its scooters and its E-bikes can create a great deal of extra value and efficiency where module commonality between their e-bicycles and scooter can allow for shared repairs and increase reuse.

4.7. Validation

A validation session was held where the group presented the findings of this project and what implications it could have for hardware specialists at Voi. The presentation can be found in Appendix 6. The main outtake from the feedback after the presentation was that modularity had been looked at and was desired by Voi for the upcoming product generations.

One aspect of why the implementation of a modular architecture had been delayed earlier was due to the technological push made between the generations. The first scooters were targeted at the consumer market, while the later models have been adapted more to work for PaaS and integrating the knowledge created during the rise of the industry.

Further, they thought that the proposal was a schoolbook example on construction and operations for a business. It was also mentioned that more specific proposals for the implementation of modularization and SMED should be presented in order to allow for a transition into the future generations.

An issue that had not been considered by the work was that cities might have policies and regulations against in-field repairs.

In the following section a value-based analysis that modularity has on the stakeholders is presented.

4.8. Stakeholder analysis of modular impact on CE and profitability

This section provides a breakdown of the value impact and profitability potential of a modular product architecture and how these elements are linked to CE. While the analysis provides useful insights, it's important to note that the scoring of each aspect has are based on assumptions. There is currently no empirical data on the perceived value of modularity and impact on the environment within the micromobility industry. These data driven assumptions are based on qualitative insights gathered through unstructured interviews, internal data, and market analysis conducted at Voi.

First, aspects and their connection to 10R, profitability, and EoL are partitioned to its stage according to section 2.7. The full breakdown is presented in Table 13 and includes a short motivation for each aspect used.

Table 13, Aspects considered in the evaluation between the product architectures

Life cycle stages	Aspects	Description	10R / profitability / EoL
Material usage over lifetime	Initial materials	The initial material used to produce the product.	R0–R2 / Processing
	Spare Parts	The flow of spare-parts that goes in and out of the product during its lifetime.	R3-R7 / Operations / Product life extension
Operation affection	Time to repair	The time it takes for each repair	Profitability / Operations
	Parts commonality	The degree of shared interfaces between components across the product portfolio.	Profitability / Product Life extension
	In-field repairs	Repairs done where it can generate value.	Profitability / Operations
	Tasks	Activities by the mechanics done in the field	Profitability / Operations
	Lifetime	The time it takes for a component to become decommissioned.	R3-R7 / Product life extension
	Design for disassembly (DfD)	Design that allows for easy separation of components	R0-R9 / Design
End of Life (EoL) processing	Material Recycling Rate	The percentage of material that can be recycled.	R8-R9 / EoL disposal

	Material separation	Separation of components that allows for materials to be recycled.	R3-R9 / Design
Other considerations	Initial Product architecture design	Determines the structure and layout of the current product.	Profitability
	Design time for upgrades	The complexity and effort required to implement upgrades within the existing product.	Profitability

Stakeholders considered in this framework were Voi, the end customer, the cities, and the enviroment (treated as a distinct stakeholder in order to quantify GHG impacts). A comparative radar chart for each stakeholder was set up to determine the impact that modularity has on the independent stakeholders.

4.8.1. Motivation to scoring

In the following section, radar charts are illustrated to show the impact that modularity has on each individual stakeholder. The scoring is comparing the value impact of the modular architecture in comparison to the current product architecture. Consequently, if the weighting is 0 this means that the value aspects remain the same as the current products architecture.

Voi

Initial Material (-1)

Voi gets affected by the addition of materials from making interfaces due to the additional couplings and wires (see Figure 7-9) that go through the product. This in turn affects their scope 3 emissions in the beginning but also works as an enabler in the long run for other aspects.

Spare parts (+3)

Enables repairs and reuse of modules, reducing the need to acquire new parts and reducing the climate impact. As mentioned in section 4.1.2 the mechanics at Voi have expressed that they don't have time to make refurbishments and reuse. With the modular operations these activities will be incorporated within the repair process.

Time to repair (+3)

With a modular product the warehouse setup can change in the way repairs are done. Instead of doing repairs for an entire scooter, the modular system allows maintenance to be done on separate modules as stated by Erixon et. al. (1996). Moreover, as expressed by Erixon et. al. (1996), and in section 4.4.2, (Figure 12) modularity can enable a parallel repair flow reducing repair time.

Parts commonality (+3)

With parts commonality Voi can increase the reuse of spare parts and share them over their entire product portfolio. This further allows the article numbers for parts for multiple products to be reduced and also reduces the inventory in Voi's local warehouses. ElMaraghy & AlGeddawy (2014) also adds that common modules can increase operational efficiency and reduce product complexity.

In field repairs (+3)

With field repairs Voi can make quick repairs in the form of modules swaps in the field as seen in Figure 10 following the SMED methodology. This reduces downtime of the scooters and allows Voi to retain the value of the scooter in field, rather than bringing them back to the warehouse for repair.

Tasks (+1)

Reduces the task complexity by removing tasks that are unnecessary for Voi. The total number of tasks is reduced, leading to a lower downtime of the scooters and less transport as proposed in section 4.4.1. This allows Voi to spend less money on unnecessary tasks and transport, increasing their profit margin.

Lifetime (+3)

As shown in 4.4.2, modularity allows the lifetime to measure per module instead of per scooter. It further enables targeted innovation and improved longevity for each individual module. Additionally, by integrating processes that allow for refurbish, reuse and repairs within the warehouse operation, each component within the module has a longer lifetime.

As shown in 4.4.2, modules are maintained and upgraded individually. Instead of decommissioning an entire scooter, only failed modules are removed. This supports R4–R6 (Repair, Refurbish, Remanufacture) and adheres to ISO 59004's long-term lifecycle thinking (Table 2)

Design for Disassembly (+3)

The proposed modular interfaces (Figure 10-11) simplify DfD by reducing the dependency of adjacent components. With a modular architecture it facilitates easy separation of components for repair, refurbishment, and recycling. This allows for faster repair cycles and the reuse ability of spare-parts in the product. Additionally, it improves the enviroment and operational efficiency within the warehouse and the fieldwork.

Material recycling rate (+2)

With continuous upgrades of material selection in the modules over a scooter lifetime, the recycling rate of the materials can increase, further adding to the EoL (recovered resources in the recycling process).

Material separation (+2)

Supports effective separation and sorting of materials, improving the recycling process. Consequently, modules allow for seamless separation and enable cleaning of different materials. As noted in 4.1.2 *Observations from Product Disassembly and Employee Input* this is currently a struggle for Voi. Improving this process would also improve the supported theories of DfR and DfD in section 2.4.2 – 2.4.3.

Initial product architecture (0)

The upfront costs to develop the modular platform are mostly put on the developer of the product. As a result, Voi does not have to make any adjustment to go from the initial product architecture to the modular one. Consequently, Voi is considered not affected.

Design time for upgrades (+2)

Enables incremental upgrades at a module level. Voi can make new innovations to modules without having to quire new units. This aligns with Baldwin & Clark (1997) and modular upgradeability benefits discussed in Mignacca et al (2025)

In Figure 18, the value effect for Voi is shown in a radar graph. There it can be seen that switching product platforms would have an effect on almost the entire value creation for Voi during its lifetime.

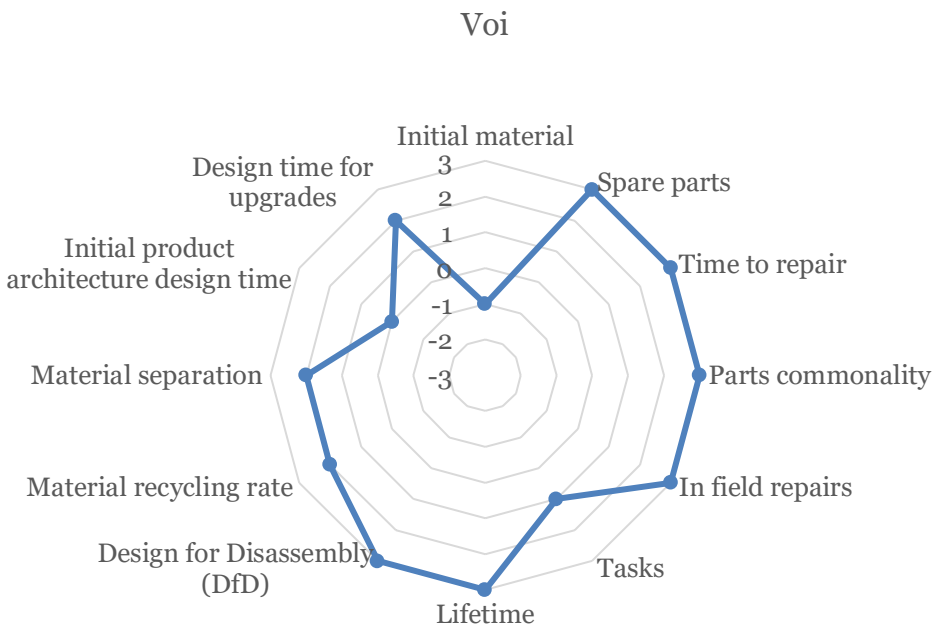


Figure 18, Radar graph of the effect on the value for each aspect concerning Voi

Ninebot

Initial Material (-2)

By using a modular system, there is a need to design the interfaces between the modules which often use more material. This is bad for Ninebot since it increases their cost for making the same product.

Spare parts (-3)

Ninebot may see reduced sales of new spare parts, limiting their incentive to support this aspect of the model due to modularity working as an enabler for greater reuse and refurbishment. This is bad for them, since one of their income sources is from selling spares.

Lifetime (-2)

Regarding the lifetime extension using a modular product architecture, this affects Ninebot negatively since it can mean that their sales number decreases. An increased lifetime of the scooter and a majority of its components means that their customers would not need to purchase as many to keep the same operational size.

Parts commonality (+3)

By using modularity, certain modules can be standardized for all their products. This means that more products have common parts between them, lowering the total amount of article numbers needed to produce different products. That in turn increases the possibility to work with economies of scale for certain parts of the scooters. ElMaraghy & AlGeddawy (2014) also support this insight also discussing how a modular platform support economy of scale.

Material recycling rate (+1)

Within the markets that currently operate Ninebots commercial products, there is a trend in lowering emissions and working with recycling in order to close the loop. Using more recycled materials is therefore a selling point that they can use to get a competitive advantage. This is also shown in the modular architecture where the retention of EoL (Figure 16) becomes more concentrated.

Increasing it can at the same time mean that the material sourcing is more expensive compared to purchasing virgin materials. This therefore lowers the score somewhat.

Material separation (+1)

The same goes for material separation as for material recycling rate. By enabling a better material separation, the operators have a greater incentive for choosing their product.

Initial product architecture design time (-3)

Creating a modular product architecture from scratch is often more time and resource intensive compared to making a new product. It is therefore a negative aspect for Ninebot, who have to put quite a bit of effort into designing it. This is also mentioned by Pahl & Beitz (1996) who argues that one of the disadvantages for the manufacturer is the increase of up-front development costs for the system.

Design time for upgrades (+3)

When a new technology or solution is to be integrated into the existing platform, it becomes easier compared to designing a brand-new scooter. This is due to the interfaces between the modules that allow for only redesigning a single module instead of the whole scooter. As previously mentioned for Voi Baldwin & Clark (1997) and Mignacca et al (2025) agrees with this.

Ninebot is not concerned with the operational performance of the scooter after the cradle stage. These aspects therefore do not affect them.

Time to repair (0)

In field repairs (0)

Tasks (0)

Design for Disassembly (0)

In Figure 19, a radar diagram of the influence for Ninebot is shown. There it can be seen that most of the effect lies in the initial stages of the scooter's lifetime.

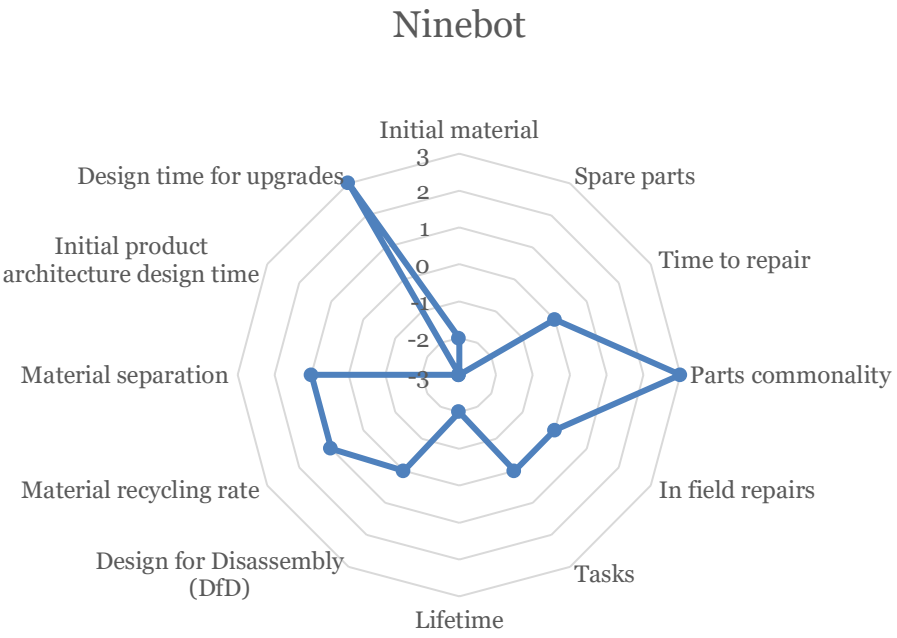


Figure 19, Radar graph of the effect on the value for each aspect concerning Ninebot

City

Time to repair (+1)

Having quicker and more effective repairs means that less broken scooters occupy the city environment. Consequently, this aspect is positive for the city.

In field repairs (-3)

With in-field repairs, it makes mechanics visible to the end users which may raise concerns for cities and the users. This further means that the streets would be occupied with mechanics making repairs, leading to a negative city effect.

Tasks (+1)

With a lower number of unnecessary tasks, a lower amount of transport can take place. This reduces the urban congestion and traffic in the city, making a positive effect.

In Figure 20, a radar diagram of the influence of the city is shown. In the radar chart it can be seen that the city is only affected positively by the decrease in tasks, while increased in-field repairs have a big negative impact.

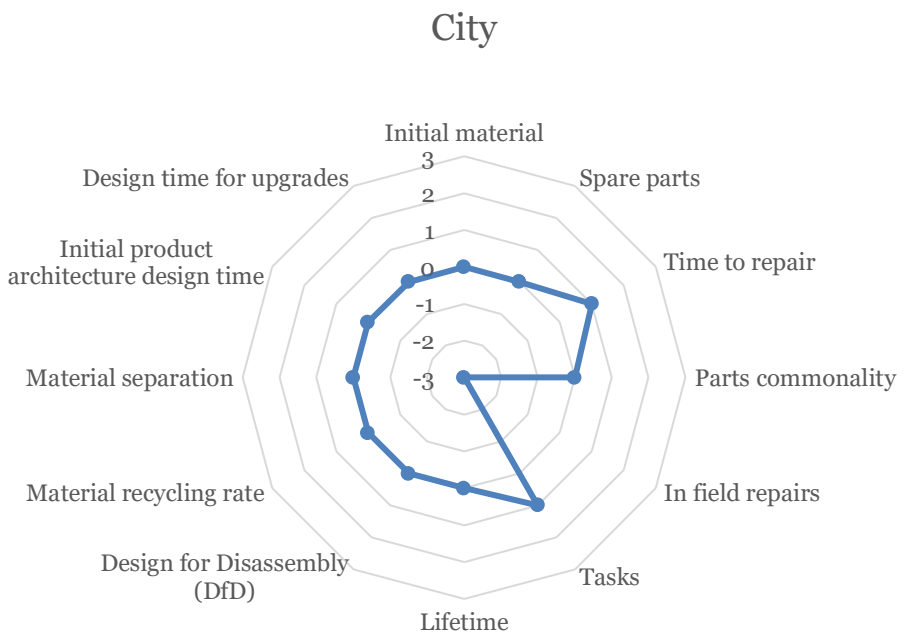


Figure 20, Radar graph of the effect on the value for each aspect concerning the city

End customer

The end customer is not affected significantly at all by the product architecture. The value of the product is much more regarding transporting one from A to destination B and that there is a scooter available to them. Consequently, all aspects set to zero are not included in the scoring motivation.

Time to repair (+2)

By decreasing the repair time, more value can be provided with the same number of scooters due to the increased uptime of the service. This therefore increases the value for the end user.

In field repairs (-1)

When mechanics are doing field repairs it occupies the streets, which may concern the end customer.

In Figure 21 a radar diagram of the influence for end customer is shown. In Figure 21, a radar diagram of the influence for end customer is shown. The end customer is mostly unaffected by modularity; however, the increase of availability and functional scooter results in a positive impact.

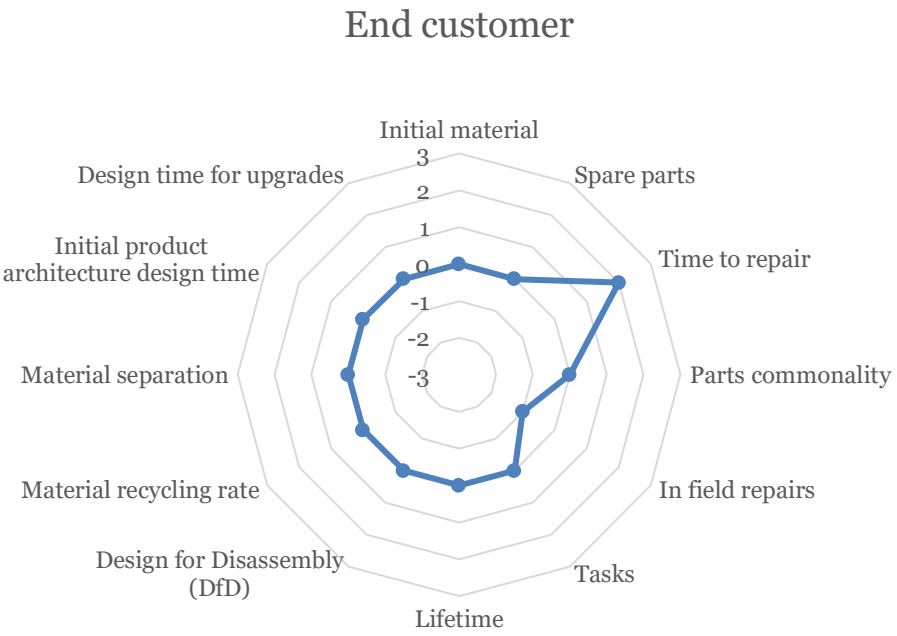


Figure 21, Radar graph of the effect on the value for each aspect concerning the end customer

Environment

Initial Material (-1)

With an increase of material usage to accommodate for the increase in interfaces a larger number of materials has to be extracted, leading to a net negative impact on GHG.

Spare parts (+2)

With standardized processes for refurbishment and repairs, more spare parts are integrated back into the resource loop and less new spare parts has to be manufactured.

Time to repair (+2)

Similarity to spare-parts, with the new warehouse process more material is integrated back into the resource loop.

Parts commonality (+2)

With a standardized product platform, the reuse aspect can be increased, and older spare-parts can be integrated into more models within the product portfolio.

In field repairs (+3)

Making field repairs reduces the need to transport the scooter between the field and the warehouse. Consequently, this reduces traffic congestion and therefore the GHG impact.

Tasks (+1)

Similarity to in-field repairs, where the reduction of unnecessary tasks reduces the need for longer transport.

Lifetime (+3)

With a longer lifetime for each individual model the need to buy new spare parts and vehicles is reduced and the net climate effect is reduced with the reduction in material extraction. This also lowers the GHG impact with the reduction of consumed materials.

Design for Disassembly (+3)

With easier separation of materials, a higher share of components can be recycled, improving the EoL outcome by enabling more efficient material recovery.

With easier separation of materials, the number of recycled materials increases, leading to an increase in EoL due to components being separated more seamless.

Material recycling rate (+3)

With continuous upgrades of materials on a modular level the overall material recycling rate can increase leading to more resources that is put back into the resource loop.

Material separation (+2)

See design for disassembly.

Initial product architecture (0)

Remains neutral since this regard alternating the design of the product which does not affect GHG.

Design for upgrades (+1)

By allowing for upgrades on module levels the product can be upgraded without the need to acquire an entire new one. Additionally, this can also allow for more efficient materials can be continuously added into the product which can increase the EoL.

In Figure 22, a radar diagram of the influence for the environment is shown. The results shows that modularity has a positive value effect over the product life cycle.

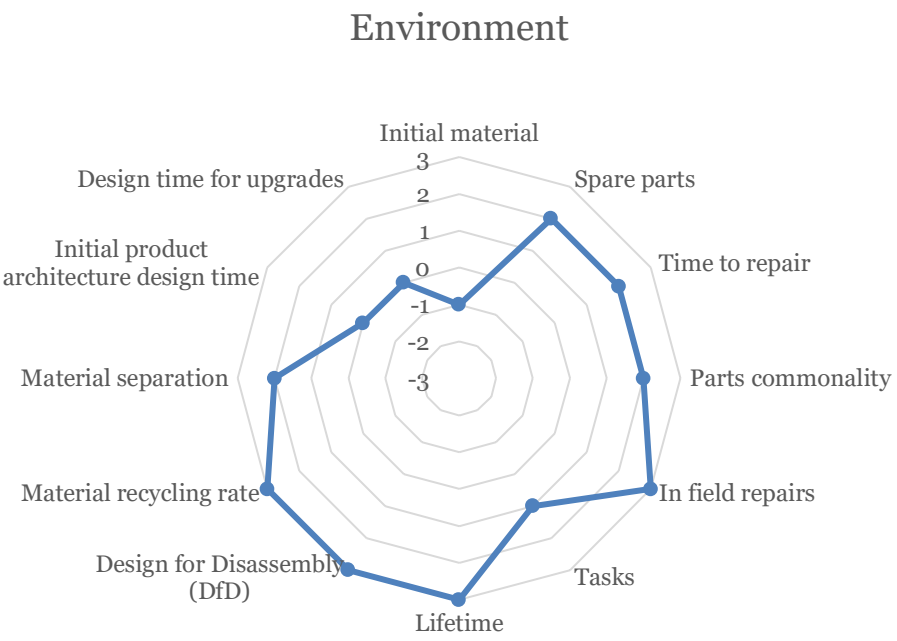


Figure 22, Radar graph of the effect on the value for each aspect concerning the environment

In Table 14 and Figure 23 the full breakdown of the result is presented. The framework reveals that the main stakeholders impacted by modularity are Voi and Ninebot. Out of the 17 analyzed aspects, the end customer was directly affected only twice, and the city three times. Voi sees costs reduction, improved operation, and environmental impact. In contrast, Ninebot bears most of the development and manufacturing cost without seeing equal returns.

Table 14, Full scoring for the different aspects from the different stakeholders.

Aspect	Ninebot	Voi	End customer	City	Environment
Initial material	-2	-1	0	0	-1
Spare parts	-3	3	0	0	2
Time to repair	0	3	2	1	2
Parts commonality	3	3	0	0	2
In field repairs	0	3	-1	-3	3
Tasks	0	1	0	1	1
Lifetime	-2	3	0	0	3
Design for Disassembly (DfD)	0	3	0	0	3
Material recycling rate	1	2	0	0	3
Material separation	1	2	0	0	2
Initial product architecture design time	-3	0	0	0	0
Design time for upgrades	3	2	0	0	0
Sum	-2	24	1	-1	20

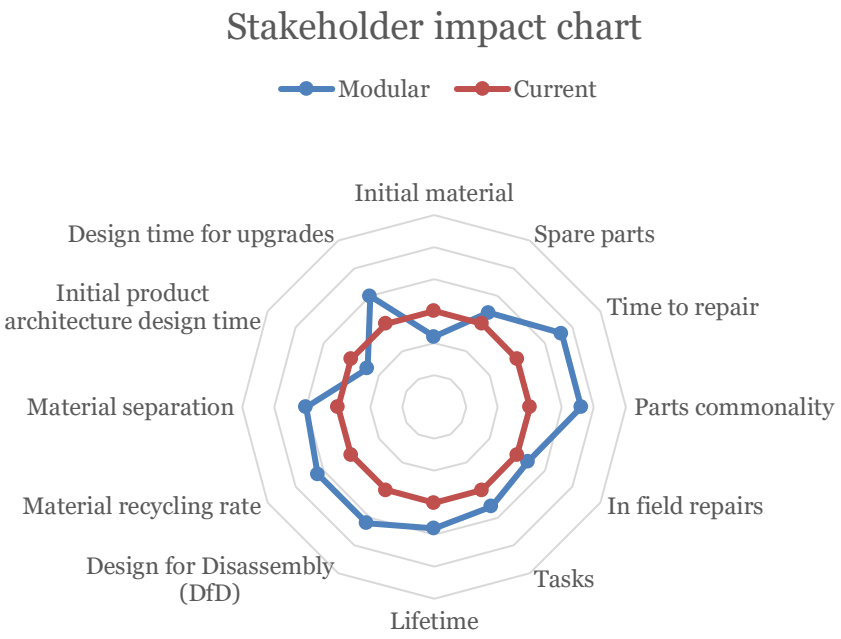


Figure 23, Modular product architecture compared to the current one.

5. Discussion

In this chapter, the role of modularity in enabling CE and profitability within a PaaS micromobility context is discussed. The analysis builds on the 10R framework (Cramer, 2020), operational efficiency based on Lean's 8 wastes, and a multi-stakeholder evaluation of value distribution. It also reflects critically on current CE practices and literature, emphasizing the importance of lifetime extension and identifying risks of greenwashing. Finally, the chapter considers methodological limitations and outlines implications for industry and future research.

5.1. Results, connection to literature and reflections

How modularity can influence the 10R framework

The 10R framework (Cramer, 2020) provides a structured way to evaluate CE strategies across a product lifecycle, ranging from product design and use (R0-R2), through lifetime extension (R3-R7), to EoL (R8-R9). Within the 10R framework, modularity offers significant potential to reinforce many of these strategies.

As shown in the results and analysis (section 4.6.1), modularity has a great effect on the R3-R7. For instance, it enables Reuse (R3) where whole modules can be reused through platform commonality across different product generations and configurations. In addition, through simplified interfaces and circular design strategies such as DfD (Prioli, et al., 2022) scooter disassembly becomes easier, and as a result the concept of Repair (R4) is also improved. Moreover, modular architecture allows for mid-life upgrades without having the acquire an entire new scooter, this further lowers the cost of making alterations to a current scooter. This aligns well with refurbishment (R5), where single modules can be upgraded during their lifecycle instead of the entire scooter. Finally, the process within the modular operations allows mechanics to consistently break the modules apart to clean, and repair the inherent component, which further enhances the process of Remanufacturing (R6). Breaking the product apparat also allows for easier handling when doing repairs and refurbishment, and as a result the repair operations also contribute to the two prior R-strategies as well.

Regarding Rethink (R1), the strategy supported further with a modular design since it encourages more use-intensive models in terms of the person km per component. This means that people riding the Voi scooter can drive a longer distance on the same components with the same resources. For Voi, this also reduces the depreciation of each component, also leading to a higher profit margin per invested resource. It further also improves the aspect of shared use between scooter models through commonality, and targeted upgrades to be able to measure each module lifetime. However, refuse (R0), as mentioned in section 4.6.1, is not directly impacted by modularity. This would require complementary approach like material substitutions, meaning that modularity would disregard materials such as plastics. Instead, modularity works as an enabler to do so, but this does not affect it directly.

Finally, the strategies between recycle and recover (R8-R9) becomes highly dependent on what materials that goes into the product and how the aspects of DfD (Prioli, et al., 2022) and DfR (Takahiro Mizuno, 2012) is considered when designing the product. Easy separation of component would allow them to be recycled, if the material allows it. Arguably, this also connects back to the first R strategies (R0-R2) (Morseletto, 2020) which concerns the earliest lifecycle stages of the product. If the product gets a design with material that can be recycled and follows the design strategies of DfD and DfR this would have a high impact on the R8-R9 strategies.

How modularity can influence the operations of a PaaS (Micromobility*)*

When the analysis was conducted at Voi it became evident that the current system included waste in transport, and overprocessing. This was particularly due to repairs being done to entire scooters instead of minor repairs. A module repair model as developed in Figure 11, enables field operations to extend which can reduce unnecessary motion and repair times. This also seems to eliminate much of the waste from today's operations.

Moreover, by coupling the modular platform with the principles of SMED (Dillion & Shingo, 1985), repairs can be allowed to shift from internal repairs to external through quick modules swaps. In the context of PaaS and micromobility this becomes highly valuable since it reduces the downtime of the scooters together with the amount of damage caused by transport handling and the amount of tasks

However, moving from internal repairs to external could affect the social sustainability due to making the work more visible for the customers. In turn, this can have a negative effect on the city as a stakeholder as seen in the value analysis in section 4.8.1. Further, making module exchanges in the streets could lead to the fleet specialists having to park their trucks for longer periods in the cities. This could become an issue since they already today have struggles finding short-time parking close to the areas where most scooters can be found.

How lifetime extension and a holistic perspective is important to reduce greenwashing while working with CE

While CE initiatives are important for driving sustainable business, the literature presented today in the subject tends to almost become a bit philosophical. With this said, many of the ideas are very good for creating businesses and offerings that are resource efficient, but in the real world, it could be seen as impossible to create something that does not consume any resources. That thought is neither in line with the first law of thermodynamics.

With that said, while conducting this work, it has been clear that the results of Stahel (2010) and Hollander et al. (2017) seem to also fall in line with this thesis regarding that lifetime extension has a much greater impact compared to recycling. When also looking at the results of this work, modularity greatly impacts just that aspect of CE, as it primarily seems to slow down the resource loop.

When conducting sales and making offerings, recycled content has almost become a buzzword for sustainability. Even if a product is made from 100% recycled content, it would still have a GHG impact since it needs to be processed to create value for a customer. Rather than making a new product of something that has already been discarded, would it not be better to instead extend the lifecycle of an already processed product to get less GHG per usage? With this said, the perspective of greenwashing should also be considered. By claiming to be a sustainable business by only using recycled content but not ensuring maximum resource retention feels a bit like misleading customers to some extent.

The holistic multi-stakeholder perspective on analyzing CE currently missing today

There is a recurring issue in the CE literature today where there is a lack of integration between circularity goals and business case around it. As argued in section 2.7.1, CE initiates must be demonstrated not only in the environmental level, but also economically. This becomes more so important in PaaS model where profitability depends directly on the asset utilization, maintenance costs, and fleet uptime. Consequently, this study proposes a novel multi-level stakeholder framework that captures CE and profitability impacts.

This fills a critical gap by enabling a more nuanced perspective on a value-based evaluation for different stakeholders. For instance, in our research we found that Ninebot bears the cost of the modular design, while Voi sees most of the benefits. This imbalance highlights why CE adoption often stalls in the supply chain, something that was noted upon by Pahl & Beitz (1996) who talks about the large upfront costs with the development of a modular platform. This is because the stakeholder value is not equal. The developed framework encourages a shard value approach, where it can become evident what hurdles external stakeholders may see, and where to build stronger incentives.

Downsides of modularity

While modularity offers several advantages in terms of circularity, operational efficiency, and scalability, it is important to acknowledge the potential drawbacks and trade-offs associated with its implementation. These downsides are both technical and systemic in nature and must be carefully considered by organizations aiming to transition toward modular product architectures.

One of the most prominent challenges is the increased complexity and cost associated with the initial design and development of modular systems. As highlighted by Pahl and Beitz (1996), modular products often

require more extensive engineering efforts to define standardized interfaces, ensure compatibility between modules, and maintain system integrity. This can lead to higher upfront development costs, longer lead times, and increased reliance on cross-functional coordination during the design phase. For manufacturers, particularly those operating in cost-sensitive markets, this may act as a barrier to adoption.

Another technical limitation concerns the physical characteristics of modular products. The need for standardized interfaces and interchangeable components can result in increased weight, volume, or material redundancy compared to fully integrated designs. This may negatively affect product performance, energy efficiency, or user experience—especially in applications where compactness and weight are critical, such as micromobility vehicles.

From a lifecycle perspective, modularity may also introduce challenges related to over-partitioning. If a product is divided into too many modules, the system may become unnecessarily complex, leading to increased maintenance requirements, higher inventory costs, and difficulties in managing spare parts. Moreover, not all components benefit equally from modularization. For parts with low failure rates or minimal upgrade potential, modularity may add cost without delivering proportional value.

There are also organizational and supply chain implications. Modularity can shift the balance of power in supplier relationships, particularly if certain modules become proprietary or are sourced from specialized vendors. This may increase dependency on specific suppliers and reduce flexibility in procurement. Additionally, if modularity is not implemented consistently across a product portfolio, it may lead to fragmentation and inefficiencies in logistics, training, and after-sales support.

Finally, modularity may face resistance from stakeholders who are not directly incentivized by its benefits. As shown in the stakeholder analysis (section 4.8), manufacturers like Ninebot may experience reduced revenue from spare parts and increased development costs, while operators like Voi capture most of the operational gains. Without mechanisms for value sharing or co-investment, such asymmetries can hinder collaboration and slow down the transition toward modular systems.

5.2. Implications for industry

The findings of this study have several important implications for companies operating within Product-as-a-Service (PaaS) models, particularly in the micromobility sector. As the results demonstrate, modular product architecture can serve as a strategic enabler for aligning CE principles with operational and financial performance. However, the transition to modularity is not without challenges, and its successful implementation requires careful consideration of organizational, technical, and systemic factors.

For operators such as Voi, modularity offers clear advantages. It enables faster and more cost-effective maintenance, reduces vehicle downtime, and facilitates reuse and refurbishment of components. These benefits translate into improved fleet availability, lower operational costs, and enhanced sustainability performance—factors that are increasingly important in competitive urban mobility markets. Moreover, modularity supports scalability by allowing companies to standardize components across different vehicle models, such as scooters and e-bikes, thereby simplifying inventory management and enabling economies of scale.

However, the study also highlights that the benefits of modularity are not evenly distributed across the value chain. While operators gain from increased efficiency and circularity, manufacturers such as Ninebot may face increased development costs, reduced spare part sales, and the need to redesign existing platforms. This asymmetry in value creation underscores the importance of establishing collaborative partnerships and shared incentive structures between manufacturers and operators. Without such alignment, the adoption of modularity may be hindered by conflicting business interests.

Another key implication concerns the design and engineering of modular interfaces. For modularity to deliver its full potential, interfaces must be robust, standardized, and compatible with rapid assembly and disassembly. This is particularly critical for safety-related systems, where regulatory compliance and user safety cannot be compromised. Companies must therefore invest in interface development and validation, potentially in collaboration with third-party testing organizations or academic institutions.

The study also suggests that internal organizational structures and performance metrics must evolve to support circular practices. For example, maintenance staff should be incentivized not only for repair speed but also for successful refurbishment and reuse of components. This requires a shift in key performance indicators (KPIs) and training programs to embed circular thinking throughout the organization.

Finally, companies must consider the broader regulatory and societal context. In-field repairs, while operationally efficient, may face resistance from municipalities due to concerns about public space usage, safety, or aesthetics. Engaging with local authorities and proactively addressing these concerns will be essential for enabling modular operations in urban environments.

5.3. Discussion of methodology

The methodological approach of this thesis was based on a qualitative case study design, centered around Voi Technology's Voiager 5 (V5) e-scooter. This approach was chosen to enable an in-depth exploration of how modular product architecture could influence CE and profitability within a micromobility Product-as-a-Service (PaaS) context. The case study method allowed for a holistic understanding of both the product and its operational environment, which was essential given the systemic nature of the research question.

One of the key strengths of the methodology lies in the triangulation of data sources. The study combined internal documentation, unstructured interviews with stakeholders across Voi's organization, direct observations at a local warehouse, and quantitative data on greenhouse gas emissions and cost structures. This multi-source approach enhanced the internal validity of the findings and allowed for a nuanced understanding of both technical and organizational aspects of modularity.

Another methodological strength was the integration of established frameworks such as the 10R model, Modular Function Deployment (MFD), and Lean's 8 wastes. These frameworks provided structure and theoretical grounding to the analysis and enabled a systematic comparison between the current and proposed product architectures. The use of stakeholder-based evaluation, where sustainability was treated as an independent stakeholder, also added depth to the analysis and allowed for a more holistic assessment of value creation.

However, the methodology also has several limitations that must be acknowledged. First, the study is conceptual in nature and does not include a physical implementation or real-world testing of the proposed modular architecture. As a result, the findings are based on theoretical modeling and assumptions, which limits their empirical generalizability. Second, the lack of access to CAD files and proprietary design data from Ninebot, the manufacturer of the V5, restricted the technical depth of the modular redesign. This meant that the proposed architecture could not be validated through simulation or prototyping.

Furthermore, the validation process was limited to internal stakeholders at Voi. While their feedback was valuable, the absence of input from Ninebot, city authorities, and end users means that important perspectives may have been overlooked. This is particularly relevant given that the stakeholder analysis revealed asymmetries in value distribution, where Voi benefits more from modularity than Ninebot. Including these external perspectives could have strengthened the credibility and relevance of the conclusions.

Another limitation concerns the use of unstructured interviews. While this format allowed for open and exploratory conversations, it also introduced potential biases and inconsistencies in data collection. Most interviews were not recorded, which may have affected the accuracy of the interpretations. Additionally, the study relied on internal data from Voi, which could not be independently verified due to confidentiality constraints.

5.4. Further research

While this thesis provides a conceptual and analytical foundation for understanding how modular product architecture can support CE and profitability in micromobility Product-as-a-Service (PaaS) operations, several areas remain open for further investigation. These areas are essential for validating, refining, and scaling the findings presented in this study.

First and foremost, future research should focus on the empirical implementation of modular product architectures in real-world settings. This includes prototyping modular interfaces, testing them under operational conditions, and evaluating their performance in terms of durability, safety, and ease of maintenance. Such studies would provide valuable insights into the practical feasibility of modularity and help identify unforeseen technical or logistical challenges.

Secondly, a full life cycle assessment (LCA) of both the current and modular product architectures should be conducted. While this thesis considered greenhouse gas emissions from material inputs, a comprehensive LCA would capture additional environmental impacts such as energy use during manufacturing, transportation emissions, and end-of-life processing. This would allow for a more robust comparison of the environmental performance of modular versus traditional systems

Another important avenue for research is the economic modeling of modularity. This includes cost-benefit analyses that account for development costs, operational savings, and potential revenue streams from refurbished or upgraded modules. Such models should also consider the distribution of costs and benefits across stakeholders, particularly in cases where manufacturers and operators have misaligned incentives.

In addition, future studies should explore user and city stakeholder perspectives on modularity. While this thesis included a stakeholder analysis, it was based on internal assumptions and qualitative insights. Surveys, interviews, or participatory workshops with end users, municipal authorities, and maintenance personnel could provide a more nuanced understanding of how modularity is perceived and what barriers or enablers exist for its broader adoption.

Furthermore, the integration of modularity with digital technologies such as IoT, predictive maintenance, and digital twins presents a promising research direction. These technologies could enhance the traceability, monitoring, and optimization of modular systems, further supporting CE goals.

Finally, the applicability of the proposed framework and findings should be tested in other PaaS sectors beyond micromobility. Industries such as consumer electronics, appliances, and industrial equipment may also benefit from modularity, but the specific design, operational, and stakeholder dynamics may differ. Comparative studies across sectors could help generalize the insights and refine the framework for broader use.

In summary, further research should aim to move from conceptual modeling to empirical validation, deepen the environmental and economic analysis, and expand the stakeholder perspective. By doing so, it will be possible to more fully understand the potential and limitations of modularity as a strategy for enabling circular and profitable business models.

6. Conclusion

This thesis sets out to investigate how modular product architecture can increase circularity and profitability within a micromobility Product-as-a-Service (PaaS) company. The research was motivated by the growing need for sustainable and resource-efficient business models, particularly in sectors with high operational intensity and environmental impact, such as shared electric scooters. Using Voi Technology's Voiager 5 (V5) as a case study, the project explored the potential of modularity to support CE strategies while maintaining or improving financial performance.

Through a structured case study approach, the current product and operational system were analyzed and compared to a conceptual modular alternative. The analysis was grounded in established frameworks such as the 10R model, Modular Function Deployment (MFD), Lean's 8 wastes, and SMED (Single Minute Exchange of Dies). A stakeholder-based evaluation framework was also developed to assess the impact of modularity on different actors, including the operator, manufacturer, city, end user, and the environment.

The results show that modularity has a strong positive influence on several CE strategies, particularly those related to product lifetime extension (R3–R7). By enabling targeted repairs, upgrades, and reuse of modules, modularity slows the resource loop and reduces the need for full product replacement. Operationally, modularity supports more efficient workflows, reduces downtime, and minimizes waste in both field and warehouse settings. These improvements translate into tangible financial benefits for the operator, while also contributing to environmental goals.

However, the study also reveals important trade-offs. The benefits of modularity are not evenly distributed across stakeholders. While Voi stands to gain significantly, Ninebot, as the manufacturer, may face increased development costs and reduced spare part sales. This asymmetry highlights the need for shared value models and collaborative innovation to ensure successful implementation. Moreover, the study identifies potential challenges related to interface design, over-partitioning, and regulatory acceptance of in-field repairs.

Methodologically, the study is conceptual and based on theoretical modeling supported by qualitative and quantitative data from Voi. While this approach provides valuable insights, it also limits the empirical generalizability of the findings. Future research should focus on prototyping, real-world testing, and broader stakeholder engagement to validate and refine the proposed concepts.

In conclusion, this thesis shows that modularity, when strategically implemented, could serve as a powerful enabler of circular and profitable business models in micromobility. By aligning product design with operational efficiency and stakeholder value, modularity offers a pathway toward more sustainable and resilient urban transport systems. The findings contribute to both academic understanding and practical decision-making and lay the groundwork for further exploration of modularity in other PaaS contexts.

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Appendix 1

To explain and clarify the meanings of 10R in this report, the analogy of a kitchen will be used to summarize each of the 10 R:s.

R0: Refuse

Analogy: Chooses to not buy a new kitchen with unnecessary accessories

Example: Decide not to buy any toxic materials in term of glues or paints when assembling your kitchen.

R1: Rethink

Analogy: Designs the kitchen in a smarter way and allow for multi-functional use.

Example: Chooses a modular system that can be adapted over time instead of switching out the entire kitchen to make changes.

R2: Reduce

Analogy: Minimizes the resource usage by buying less or more sustainable kitchen components

Example: Picking cabinets that are thinner but maintains the same durability in order to minimize the extraction of raw materials.

R3: Reuse

Analogy: Uses the same kitchen cabinets again without changing them

Example: Uses the cabinet doors from an old kitchen in a new one or uses an old tabletop in another room.

R4: Repair

Analogy: Fixes damages instead of buying new products.

Example: Fixes a broken drawer or exchange a hinge instead of buying a new cabinet.

R5: Refurbish

Analogy: Improves the kitchen by making small tweaks instead of replacing everything

Example: Paints the cabinet doors, change the handles or polish and oil the tabletop to get a new feeling.

R6: Remanufacture

Analogy: Restoring the old components of the kitchen by rebuilding them.

Example: Use old frames and build them into new cabinets by exchanging certain parts but having the same foundational structure.

R7: Repurpose

Analogy: Using kitchen components in a totally different way than what they were supposed to be used for

Example: Using old cabinet doors as a headboard or using a tabletop as a desk,

R8: Recycle

Analogy: Process the material so that it can be used to create new products.

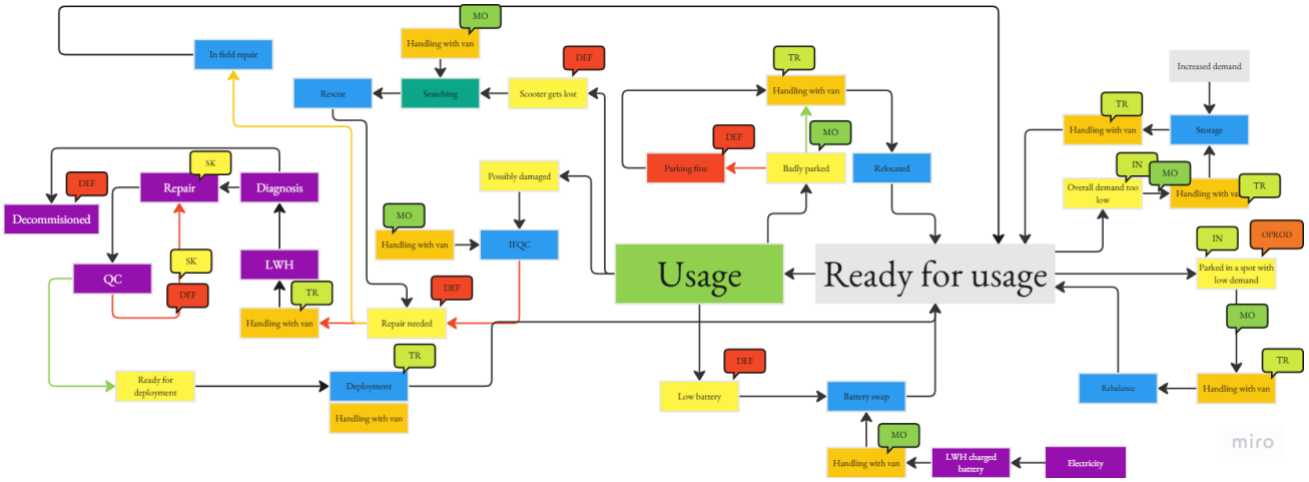
Example: Recycling an old tabletop of wood so that it can be grinded and used as raw material for Medium Density Fiberboards.

R9: Recover

Analogy: If nothing else works, extracting the material from the kitchen by combustion.

Example: Combusting the materials that cannot be used and using the heat for district heating.

Appendix 3


MO
Handling with van
In field repair
Rescue
Searching
DEF
Scooter gets lost
TR
Handling with van
DEF
Parking fine
MO
Badly parked
Relocated
Increased demand
TR
Handling with van
Storage
IN
MO
Overall demand too low
Handling with van
TR
SK
DEF
Repair
Diagnosis
Decommisioned
Possibly damaged
MO
Handling with van
IFQC
Usage
Ready for usage
IN
OPROD
Parked in a spot with low demand
SK
QC
DEF
LWH
TR
Handling with van
DEF
Repair needed
MO
TR
Handling with van
Ready for deployment
TR
Deployment
Handling with van
DEF
Low battery
Battery swap
Rebalance
TR
Handling with van
MO
Handling with van
LWH charged battery
Electricity
miro

Appendix 4

DSM Clustering of the V7 BOM

3.1. Parts considered having a function:

Handlebar	1
Phone holder	2
Bell	3
IOT Device	4
Control unit	5
Motor	6
Battery hatch	7
Battery lock	8
Battery	9
Throttle	10
Brake lever	11
Brake front	12
Brake rear	13
Turn lights steering	14
Turn lights switch	15
Turn lights rear	16
Main beam	17
Rear light	18
Frame	19
Vertical Tube	20
Hook	21
Front wheel	22
Front fork	23
Rear wheel	24
Kick stand	25
Front fender	26
Rear fender	27

3.2. DSM Clustering the V7 BOM considering all interfaces to have value 1, 4000 repetitions.

Cluster Member List

18-Feb-2025 15:00:56

Cluster #1

IOT Device (4)
Motor (6)
Battery hatch (7)
Battery lock (8)
Battery (9)
Turn lights rear (16)
Rear light (18)
Frame (19)
Front wheel (22)
Rear wheel (24)

Cluster #2

Handlebar (1)
Phone holder (2)
Bell (3)
Throttle (10)
Turn lights steering (14)
Turn lights switch (15)

Cluster #3

Control unit (5)
Brake lever (11)
Brake front (12)
Brake rear (13)
Main beam (17)

Cluster #4

Front fork (23)
Front fender (26)

Cluster #5

Hook (21)
Kick stand (25)

Cluster #6

Vertical Tube (20)
Rear fender (27)

3.3. DSM Clustering the V5 BOM considering interface G to have value 1 with the rest having 0, 4000 repetitions.

Cluster #1

IOT Device (4)
Motor (6)
Battery hatch (7)
Battery lock (8)
Battery (9)
Brake front (12)
Brake rear (13)
Rear light (18)
Frame (19)
Vertical Tube (20)
Rear fender (27)

Cluster #2

Handlebar (1)
Phone holder (2)
Bell (3)
Throttle (10)
Brake lever (11)
Turn lights steering (14)
Turn lights switch (15)

Cluster #3

Turn lights rear (16
Front wheel (22)
Front fork (23)
Rear wheel (24)
Front fender (26)

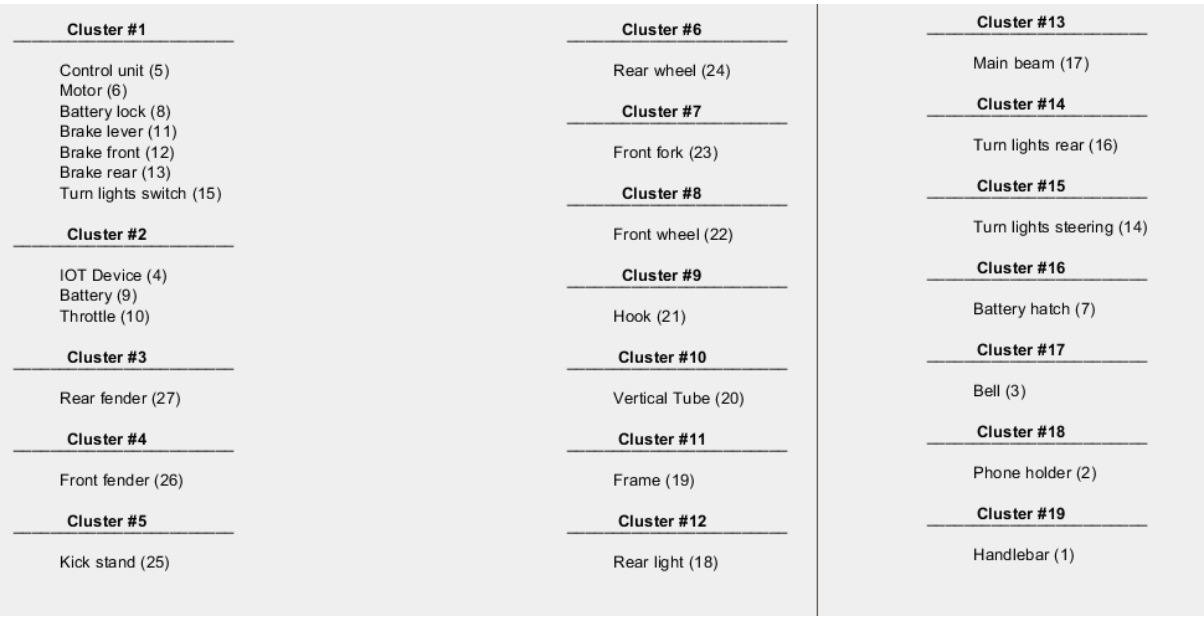
Cluster #4

Hook (21)
Kick stand (25)

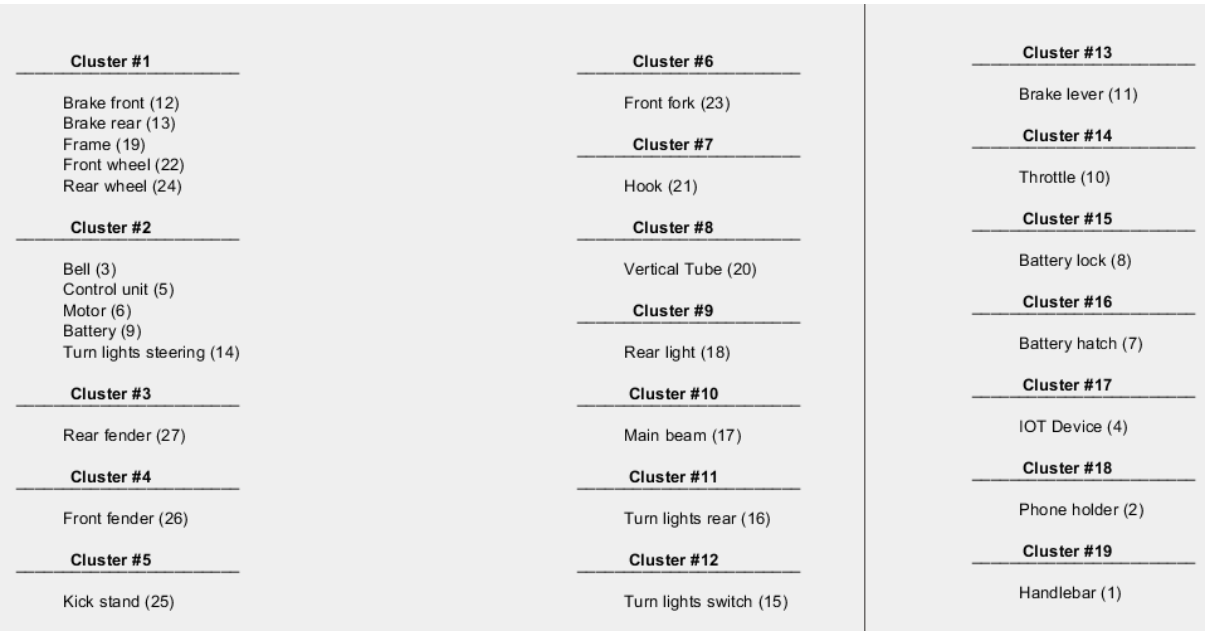
Cluster #5

Control unit (5)
Main beam (17)

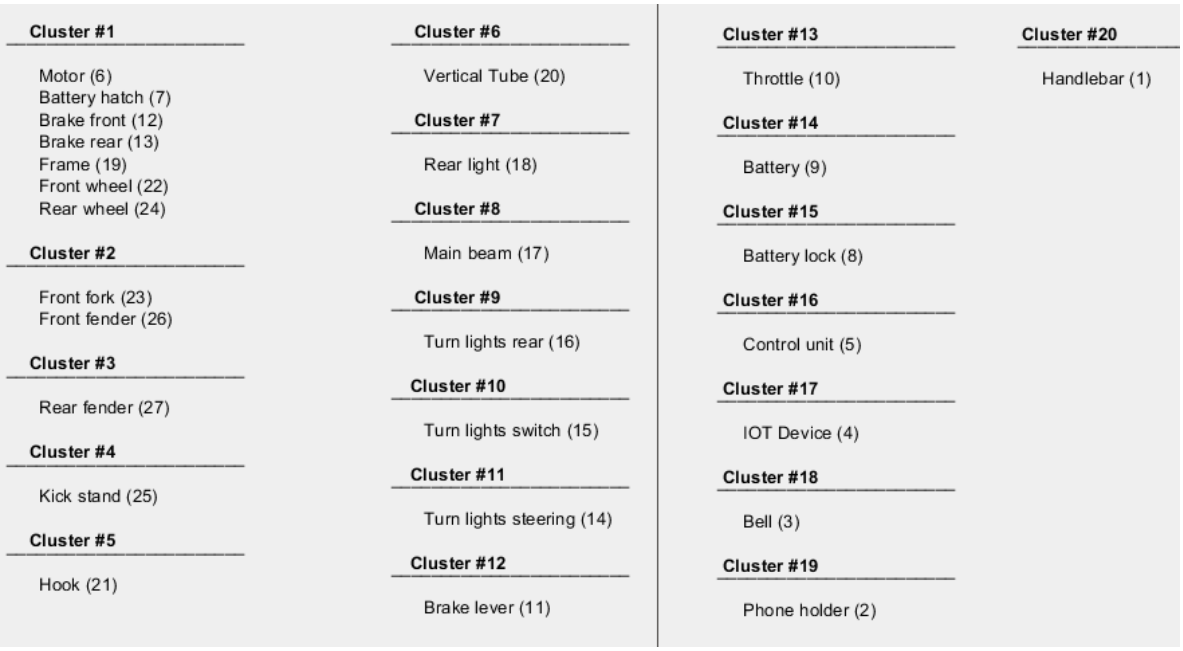
3.4. DSM Clustering the V5 BOM considering interface S to have value 1 with the rest having 0, 4000 repetitions.



3.5. DSM Clustering the V5 BOM considering interface E to have value 1 with the rest having 0, 4000 repetitions.



3.5. DSM Clustering the V5 BOM considering interface M to have value 1 with the rest having 0, 4000 repetitions.

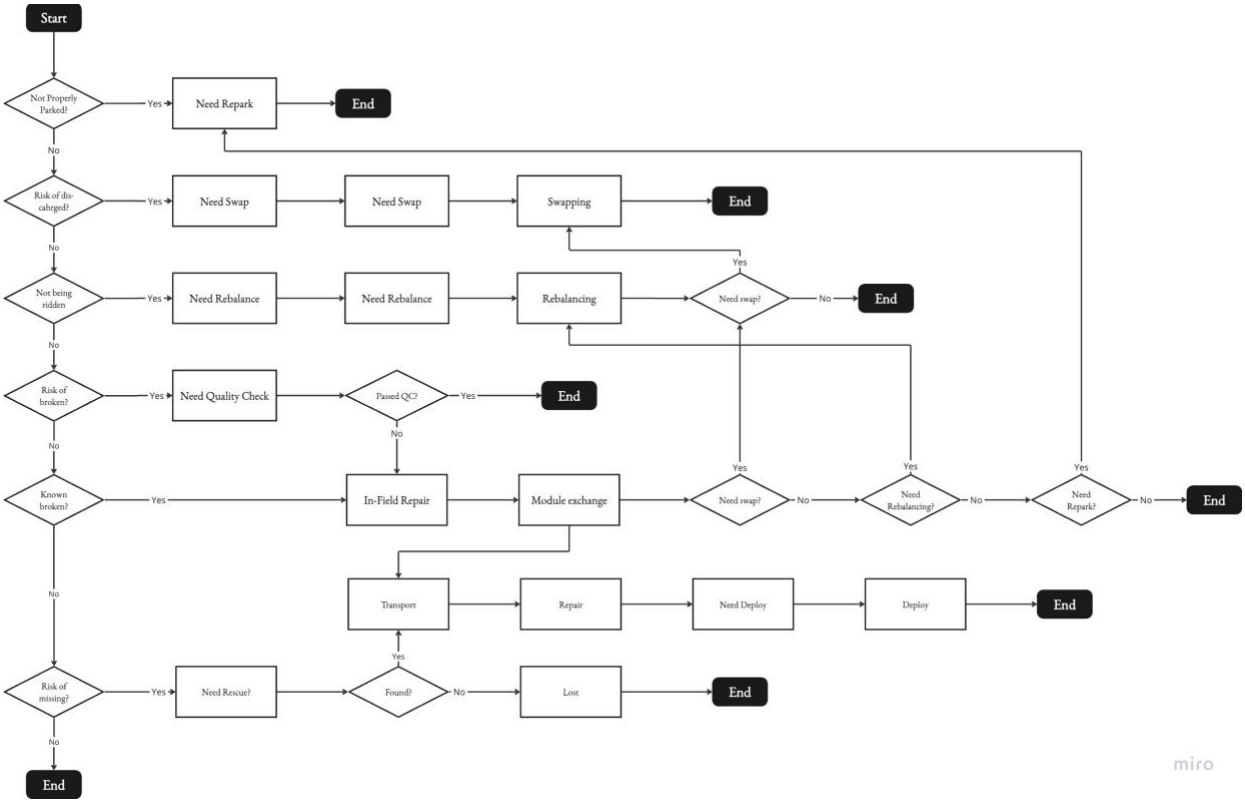


3.6. PMM table

Technical Solutions	Common unit	Process/organisation	Upgrading	Serviceability	Strategic supplier	Recycling	Separate testing	Modules
Handlebar	9							Handlebar
Phone holder	9							
Bell	9							
Throttle	9							
IOT Device	9			9			9	IOT
Control unit	9						9	Control unit
Motor	9	9		9				Propulsion
Rear wheel	9	9		9				
Brake rear	9	9		9				
Battery hatch	9							Battery hatch
Battery lock	9						9	Battery lock
Battery	9		9		9		9	Battery
Brake lever	9			9				Brake levers
Brake front	9	9		9				Front wheel
Front wheel	9	9		9				Front wheel
Turn lights steering	9					9		Turn lights steering
Turn lights switch	9					9		Turn lights switch
Main beam	9					9		Frame unit
Fork	9					9		
Frame	9					9		
Vertical Tube	9					9		
Hook	9					9		
Kick stand	9							Kickstand
Front fender	9							
Rear light	9					9		Rear fender
Turn lights rear	9					9		
Rear fender	9					9		
	243	45	9	63	9	90	36	

Module Drivers

Appendix 5


Start
Not Properly Parked?
Yes
Need Repark
End
No
Risk of dis-cahrged?
Yes
Need Swap
Need Swap
Swapping
End
No
Not being ridden
Yes
Need Rebalance
Need Rebalance
Rebalancing
Need swap?
Yes
No
End
No
Risk of broken?
Yes
Need Quality Check
Passed QC?
Yes
End
No
Known broken?
Yes
In-Field Repair
Module exchange
Need swap?
Yes
No
Need Rebalancing?
Yes
No
Need Repark?
Yes
No
End
No
Transport
Repair
Need Deploy
Deploy
End
Risk of missing?
Yes
Need Rescue?
Found?
Yes
No
Lost
End
No
End
miro

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